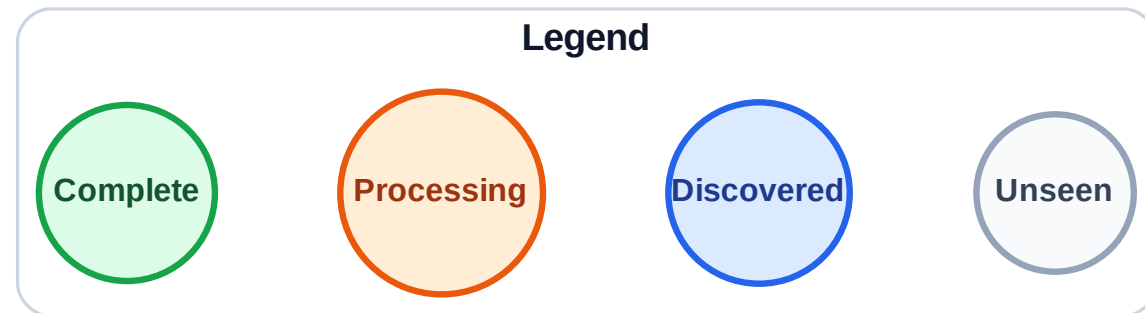
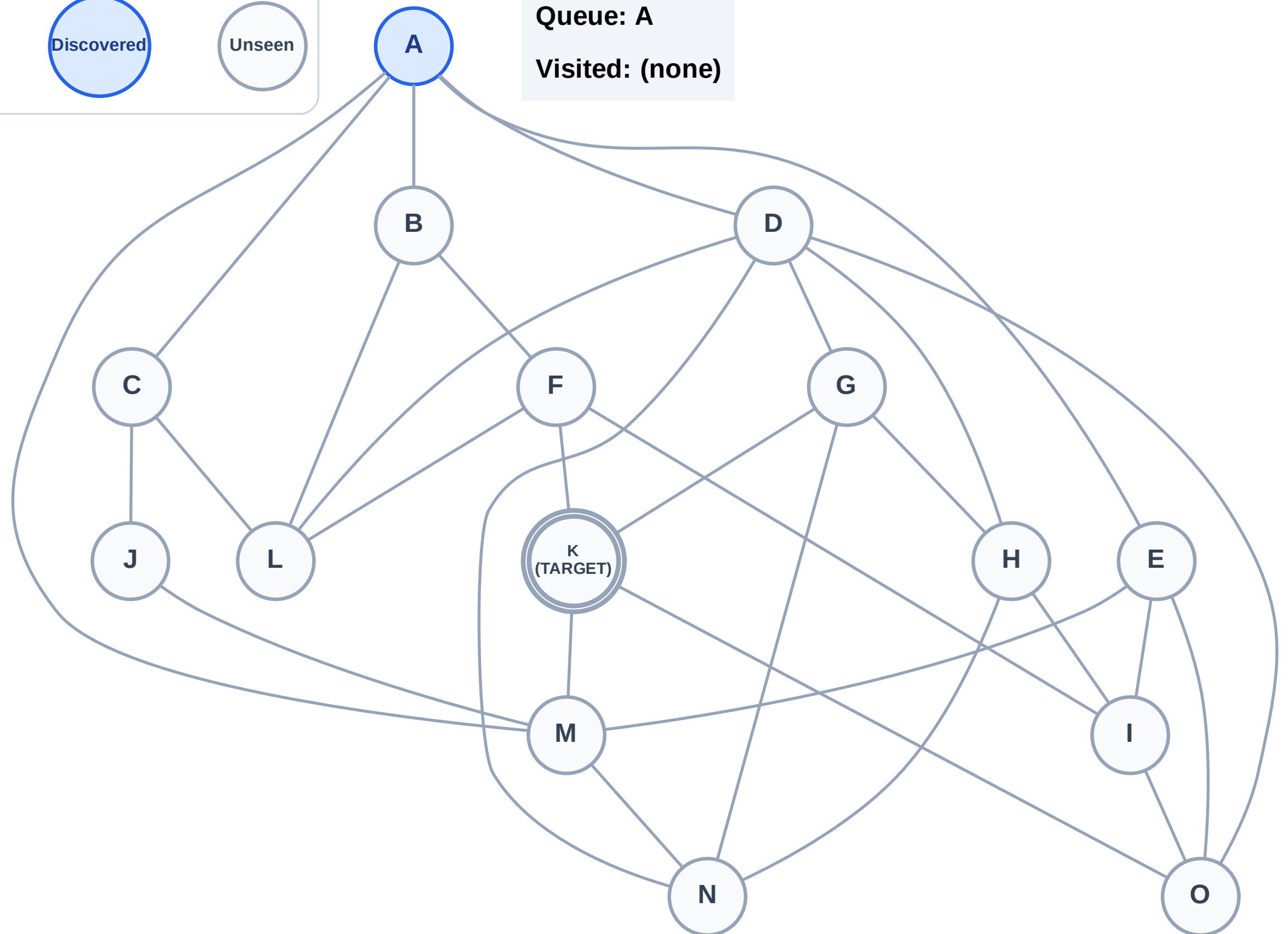


# BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A  
Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

Complete

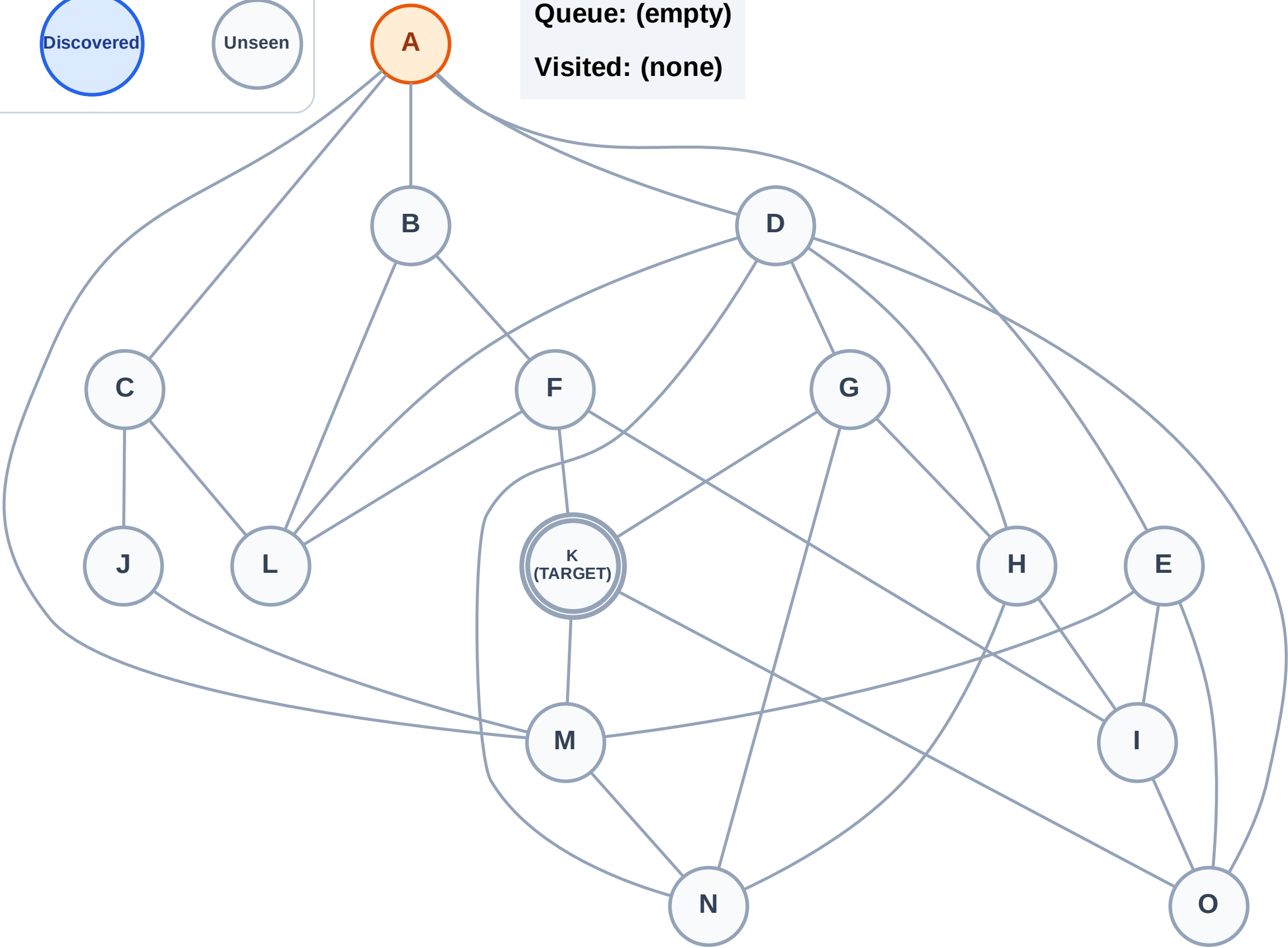
Processing

Discovered

Unseen

Queue: (empty)

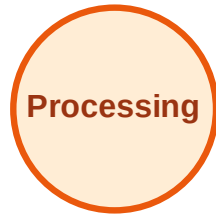
Visited: (none)



# BFS Traversal

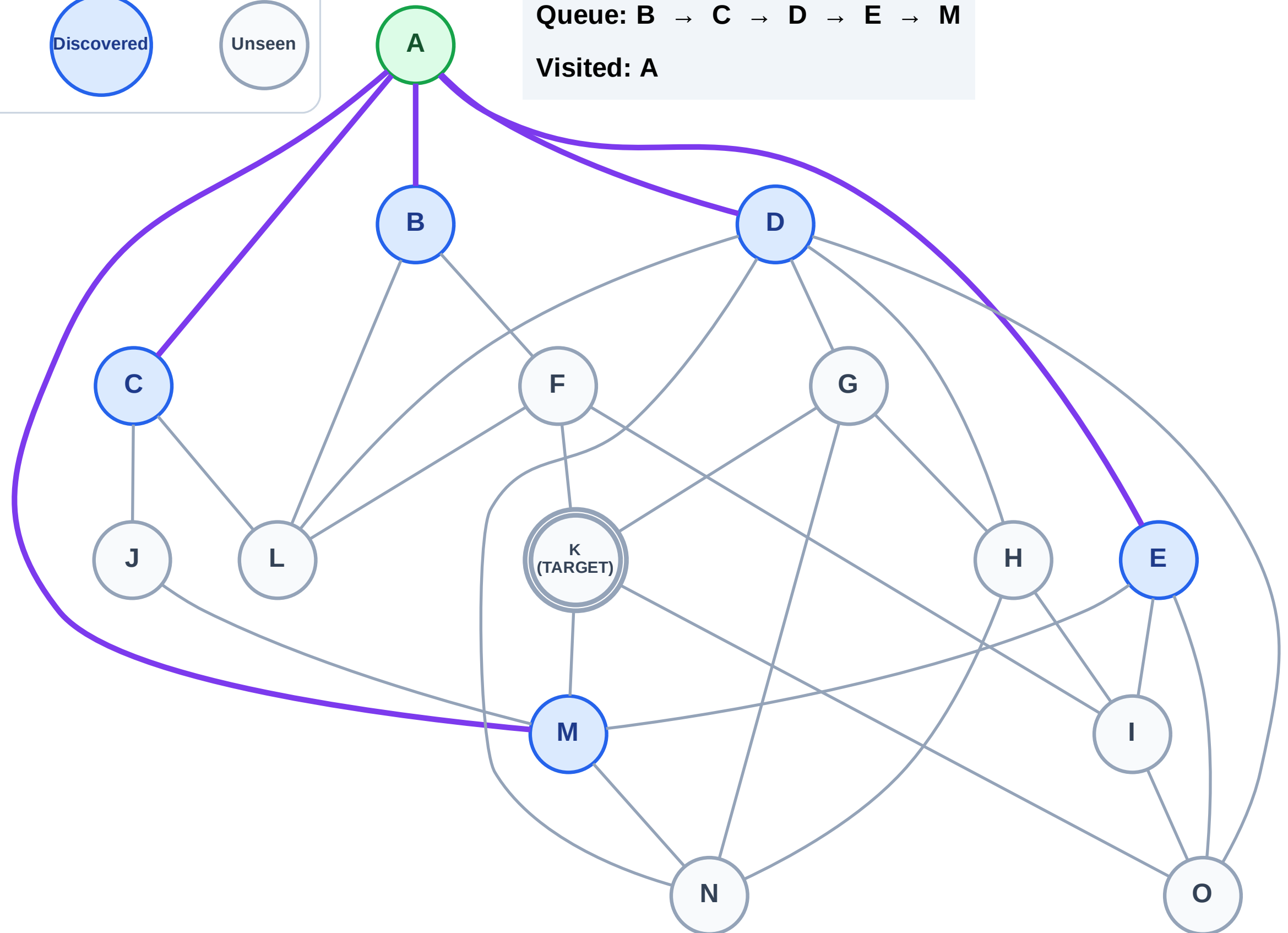
Step 3: Discover B, C, D, E, M from A

## Legend

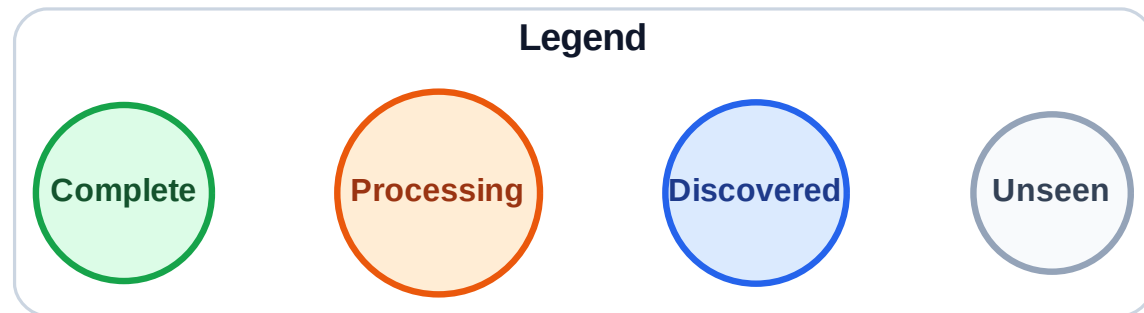


Queue: B → C → D → E → M

Visited: A

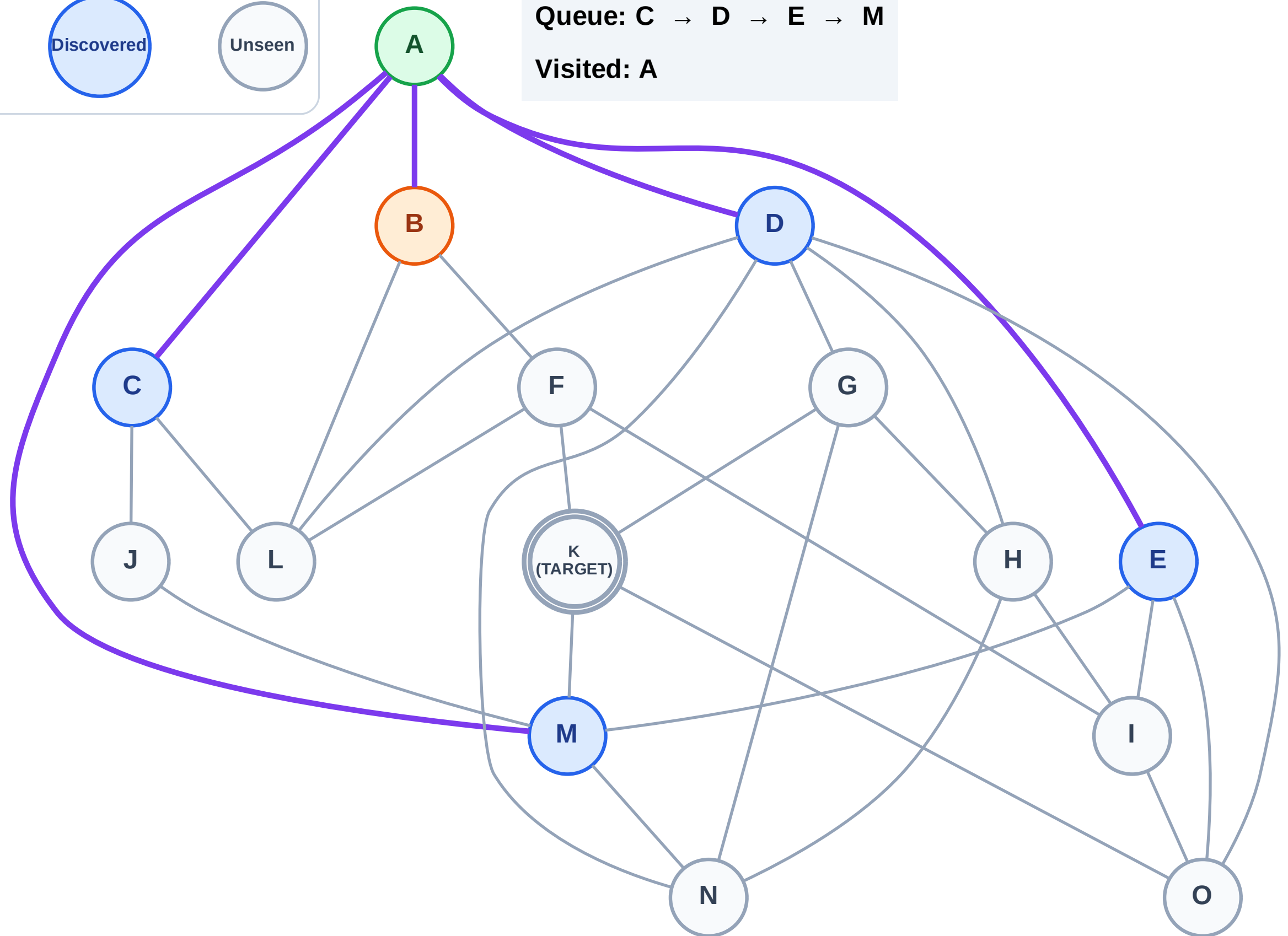


### Step 4: Process node B



**Queue: C → D → E → M**

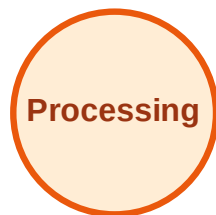
**Visited: A**



# BFS Traversal

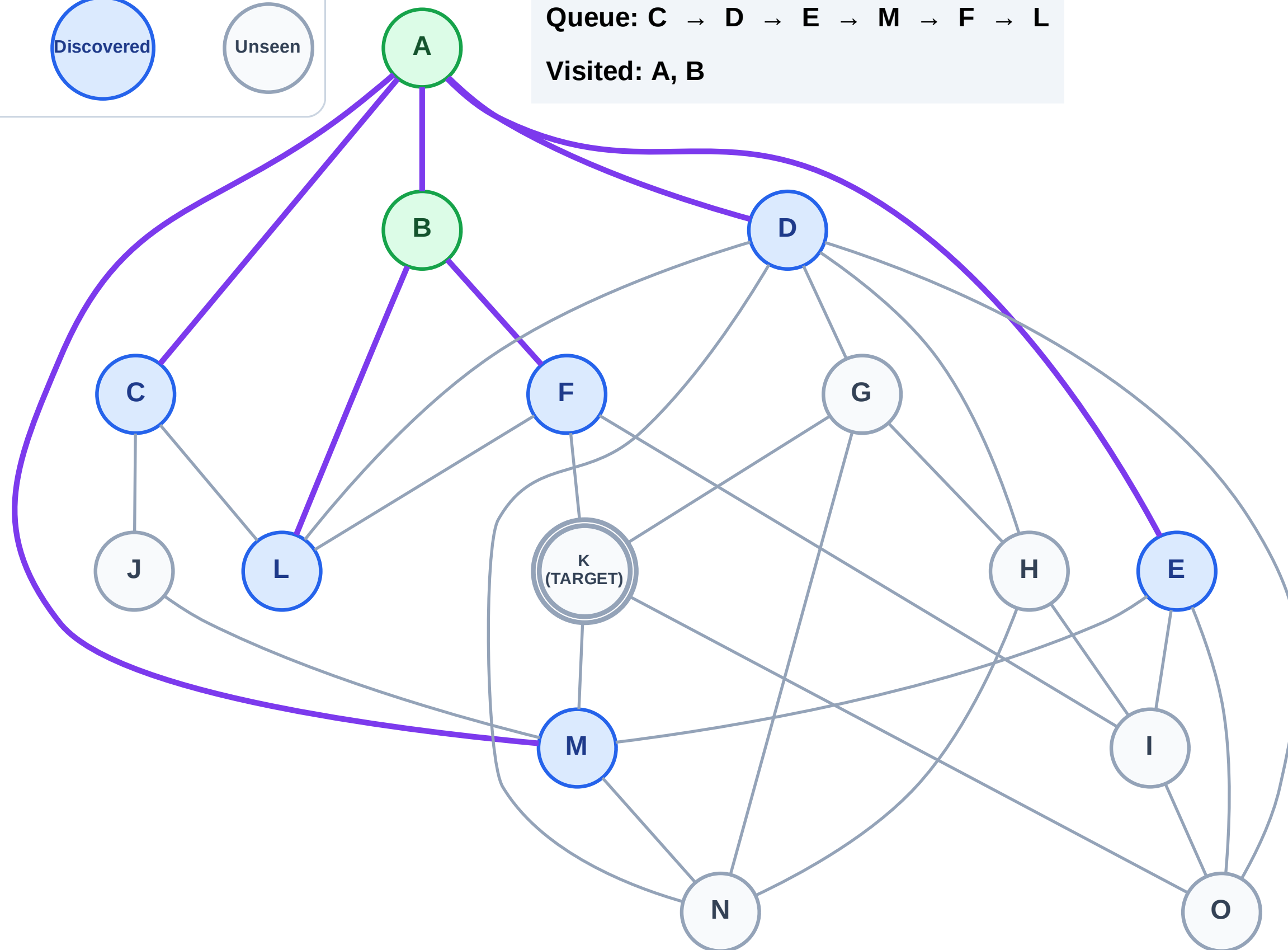
Step 5: Discover F, L from B

## Legend



Queue: C → D → E → M → F → L

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

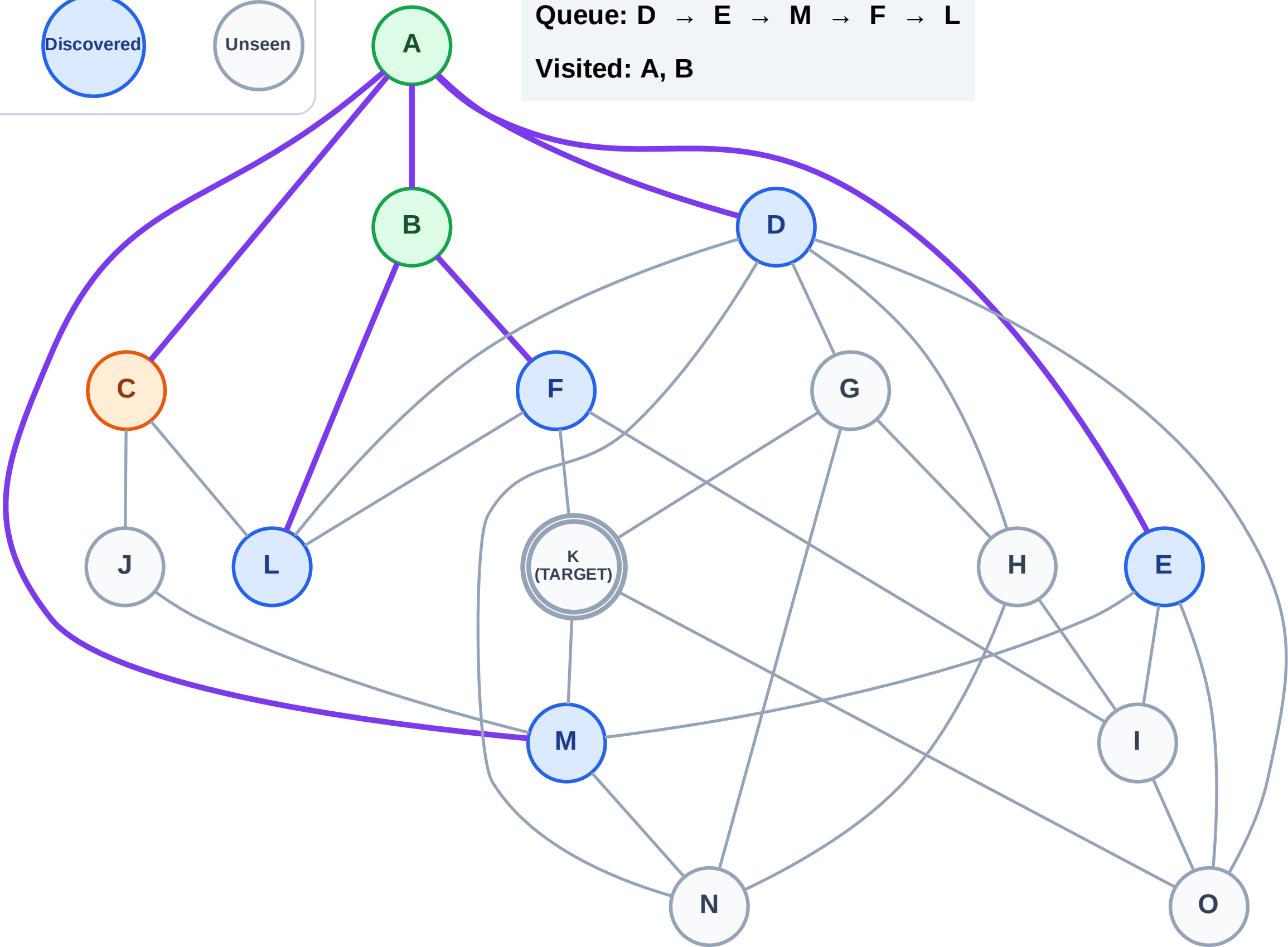
Processing

Discovered

Unseen

Queue: D → E → M → F → L

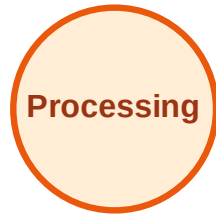
Visited: A, B



# BFS Traversal

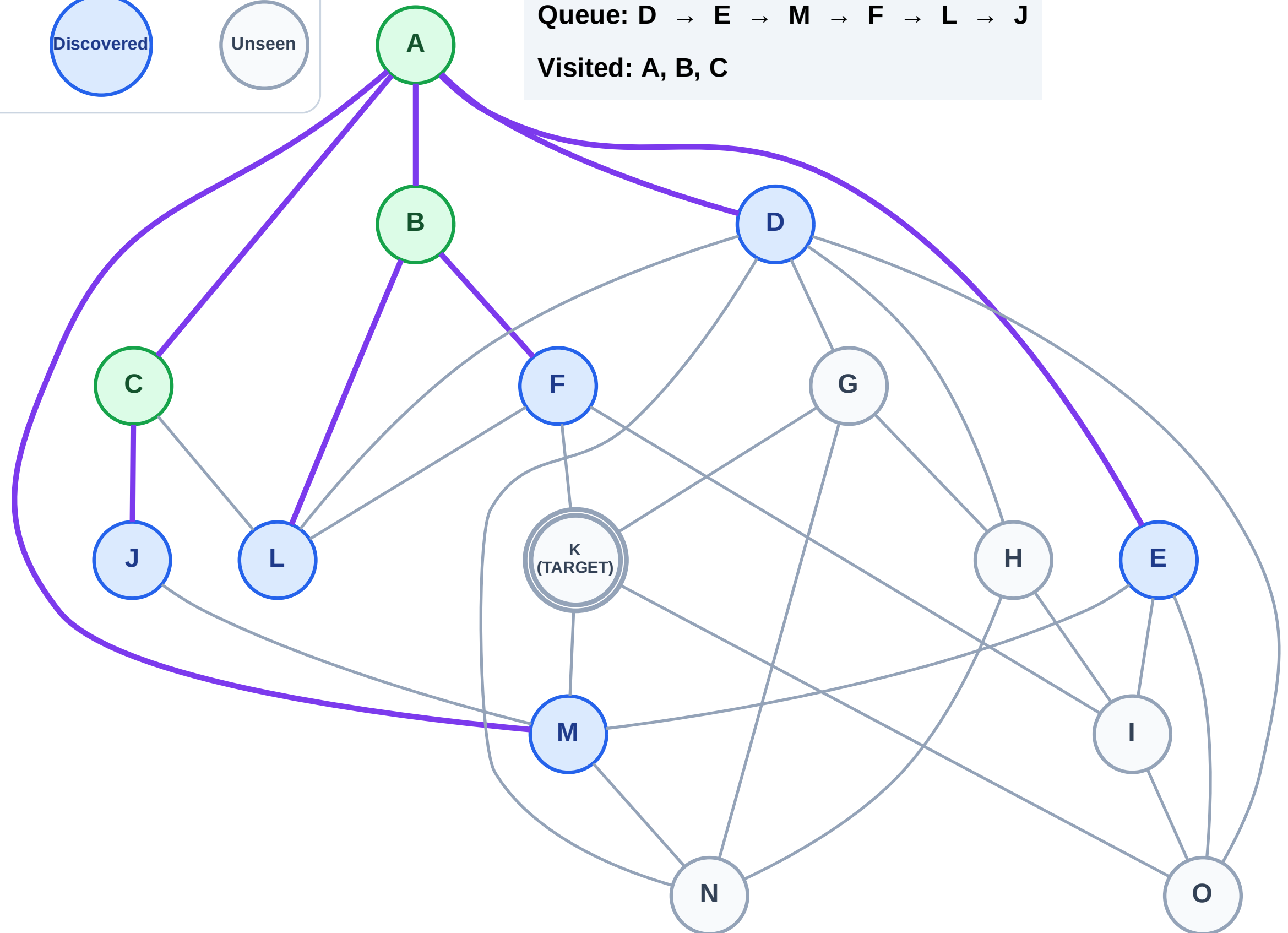
Step 7: Discover J from C

## Legend



Queue: D → E → M → F → L → J

Visited: A, B, C



# BFS Traversal

Step 8: Process node D

Legend

Complete

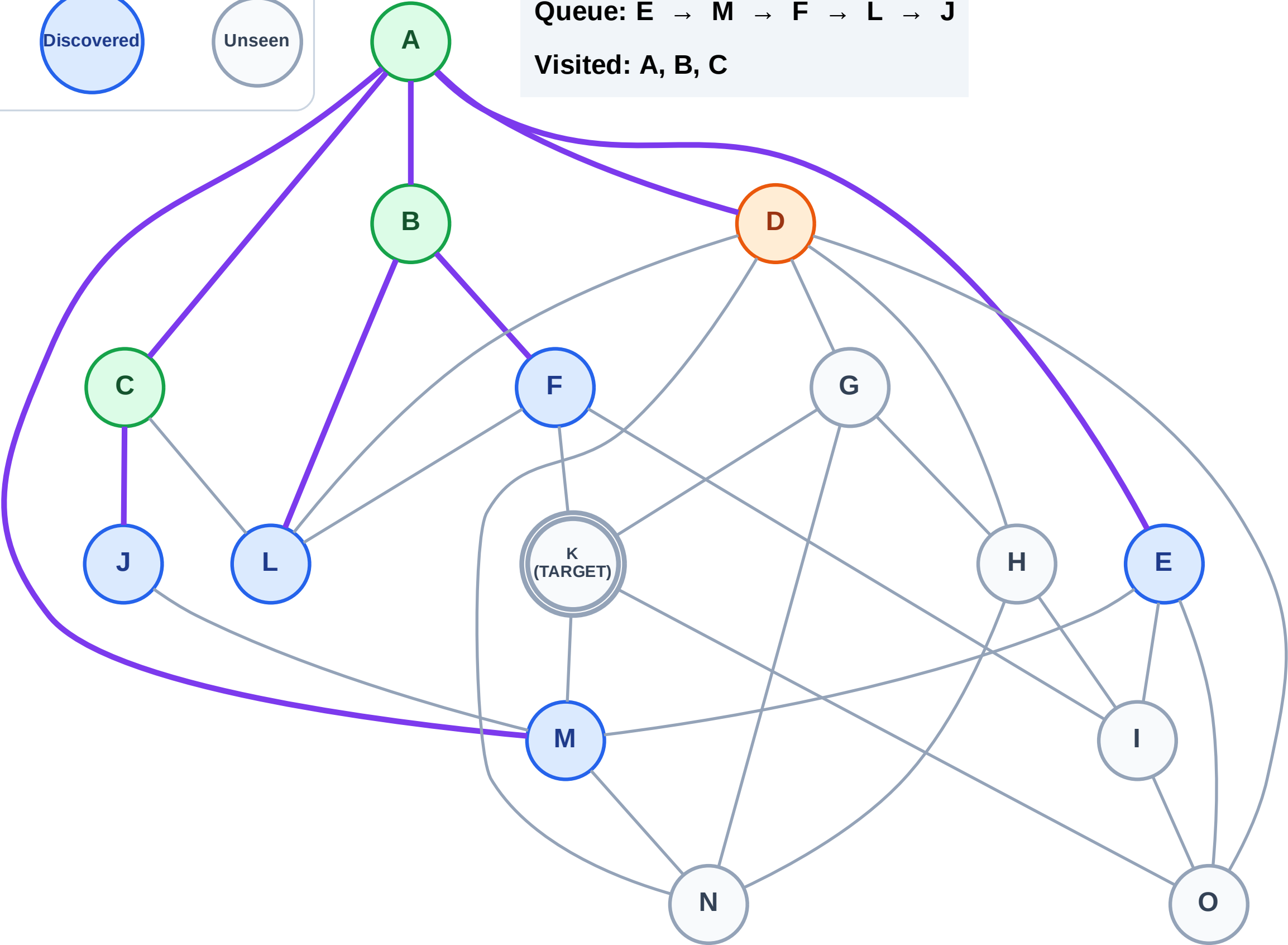
Processing

Discovered

Unseen

Queue: E → M → F → L → J

Visited: A, B, C



# BFS Traversal

Step 9: Discover G, H, N, O from D

Legend

Complete

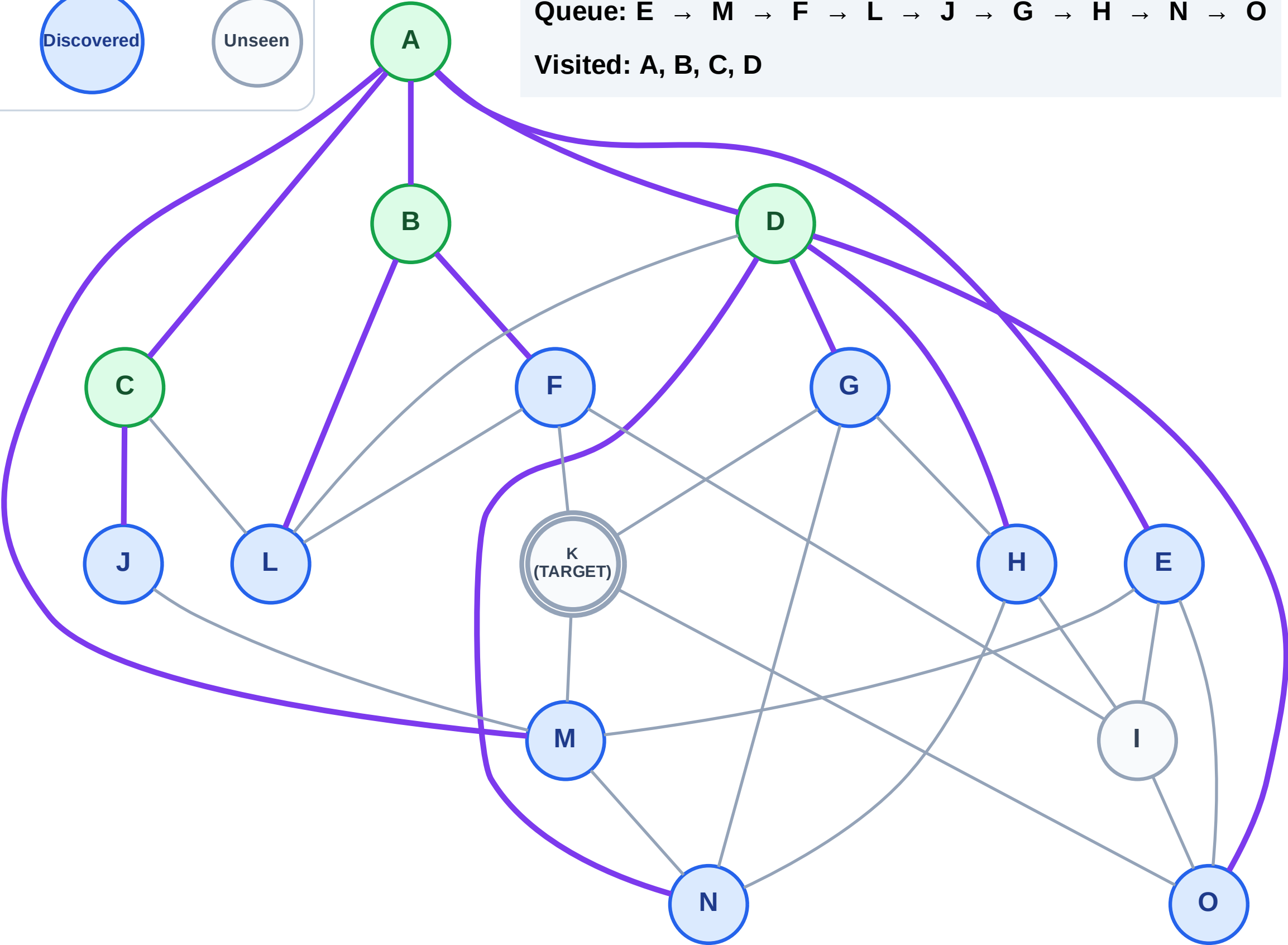
Processing

Discovered

Unseen

Queue: E → M → F → L → J → G → H → N → O

Visited: A, B, C, D



# BFS Traversal

Step 10: Process node E

Legend

Complete

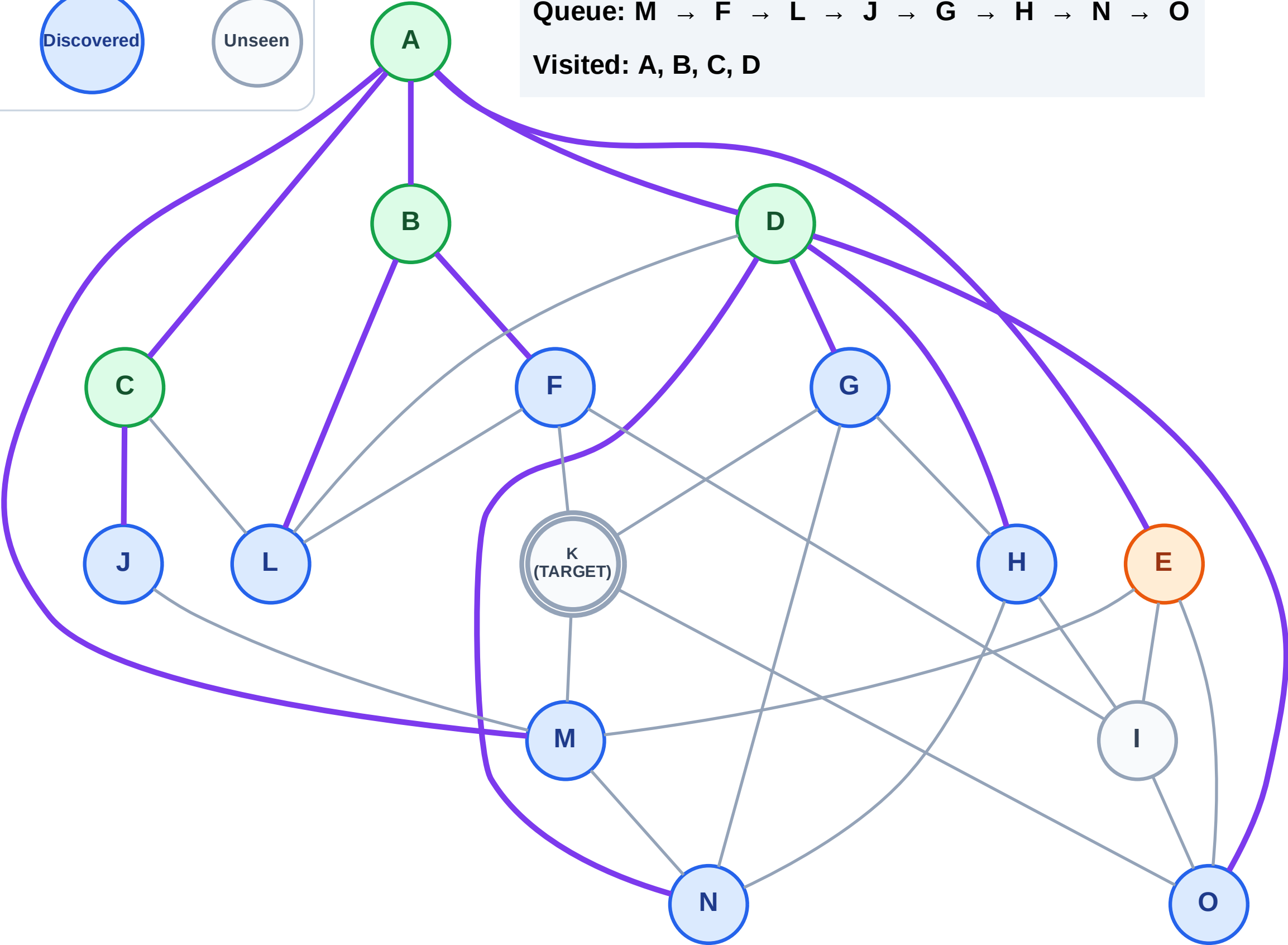
Processing

Discovered

Unseen

Queue: M → F → L → J → G → H → N → O

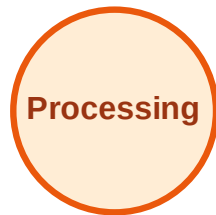
Visited: A, B, C, D



# BFS Traversal

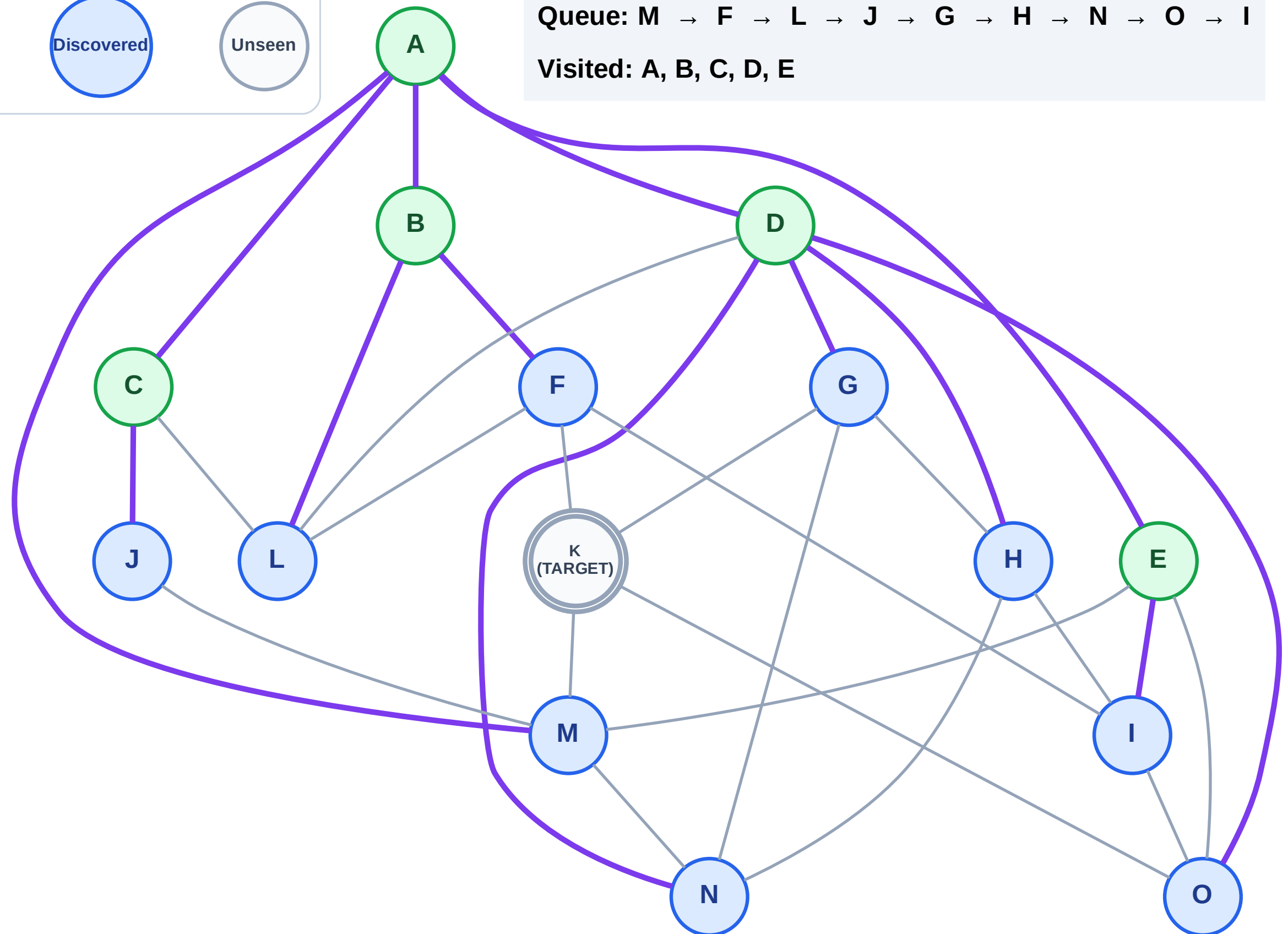
Step 11: Discover I from E

## Legend



Queue: M → F → L → J → G → H → N → O → I

Visited: A, B, C, D, E



# BFS Traversal

Step 12: Process node M

Legend

Complete

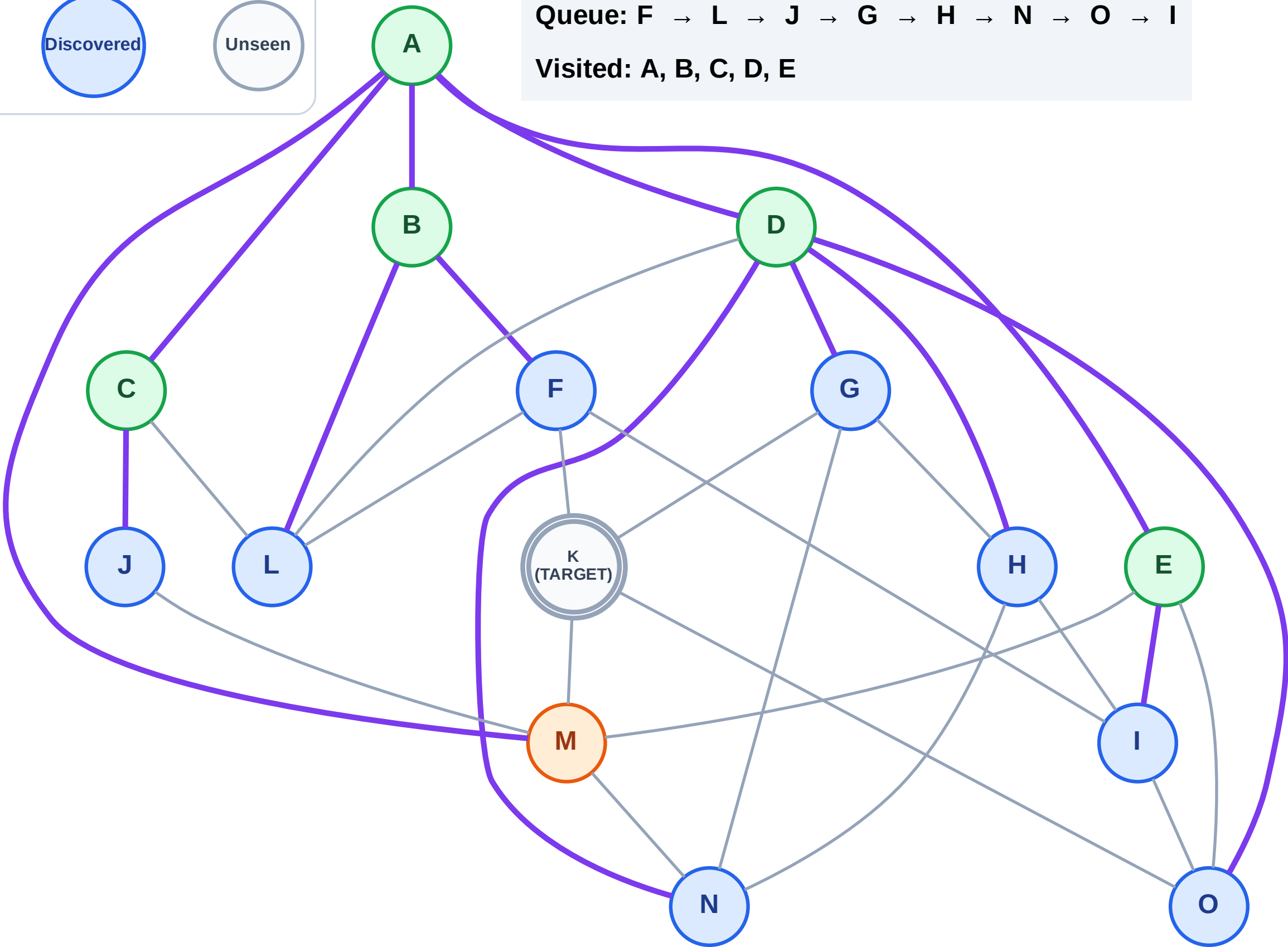
Processing

Discovered

Unseen

Queue: F → L → J → G → H → N → O → I

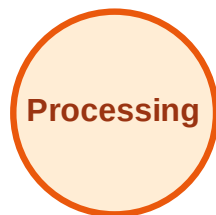
Visited: A, B, C, D, E



# BFS Traversal

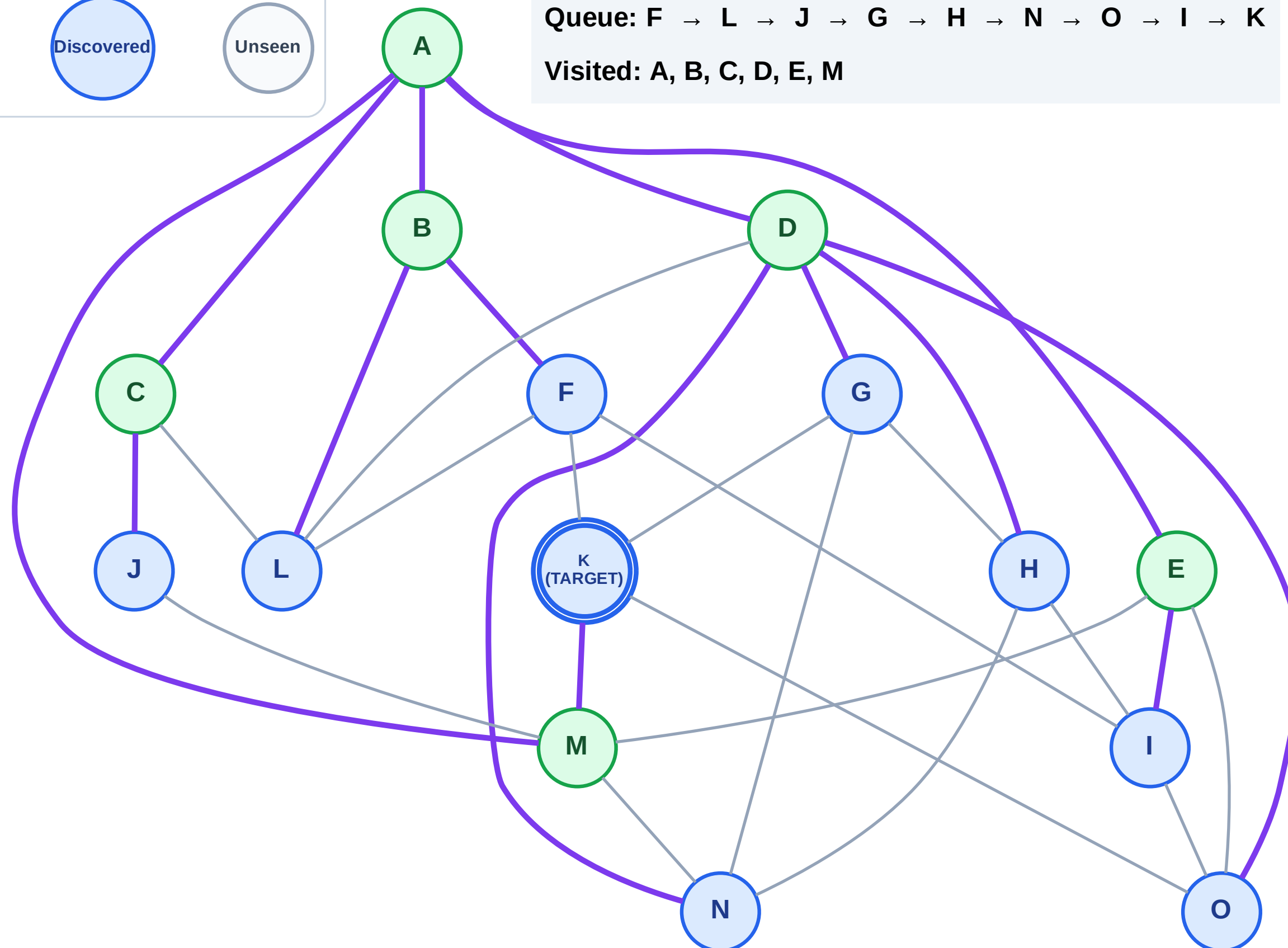
Step 13: Discover K from M

## Legend



Queue: F → L → J → G → H → N → O → I → K

Visited: A, B, C, D, E, M



# BFS Traversal

Step 14: Process node F

Legend

Complete

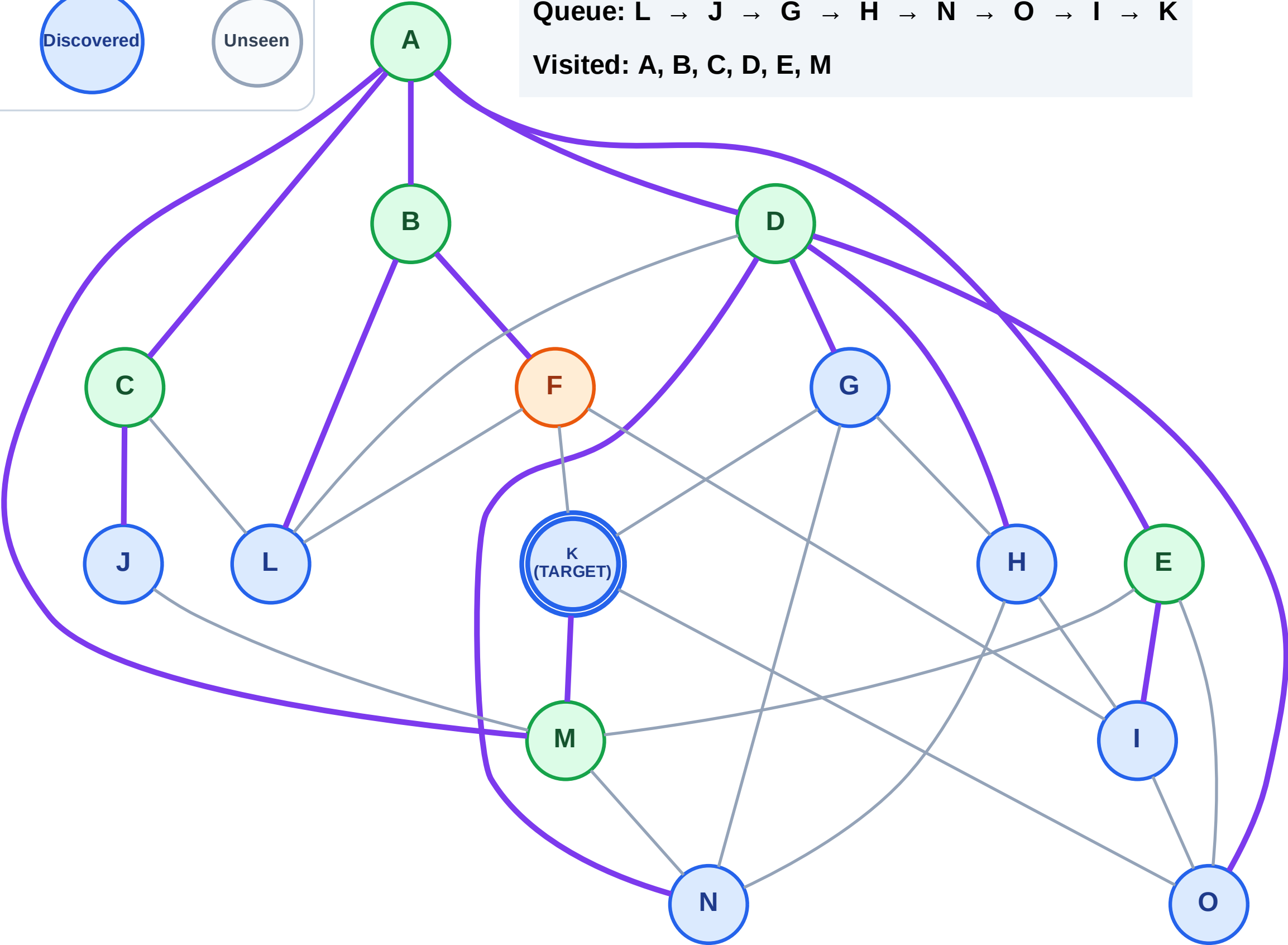
Processing

Discovered

Unseen

Queue: L → J → G → H → N → O → I → K

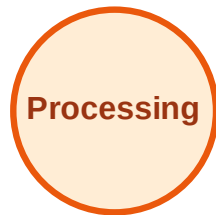
Visited: A, B, C, D, E, M



# BFS Traversal

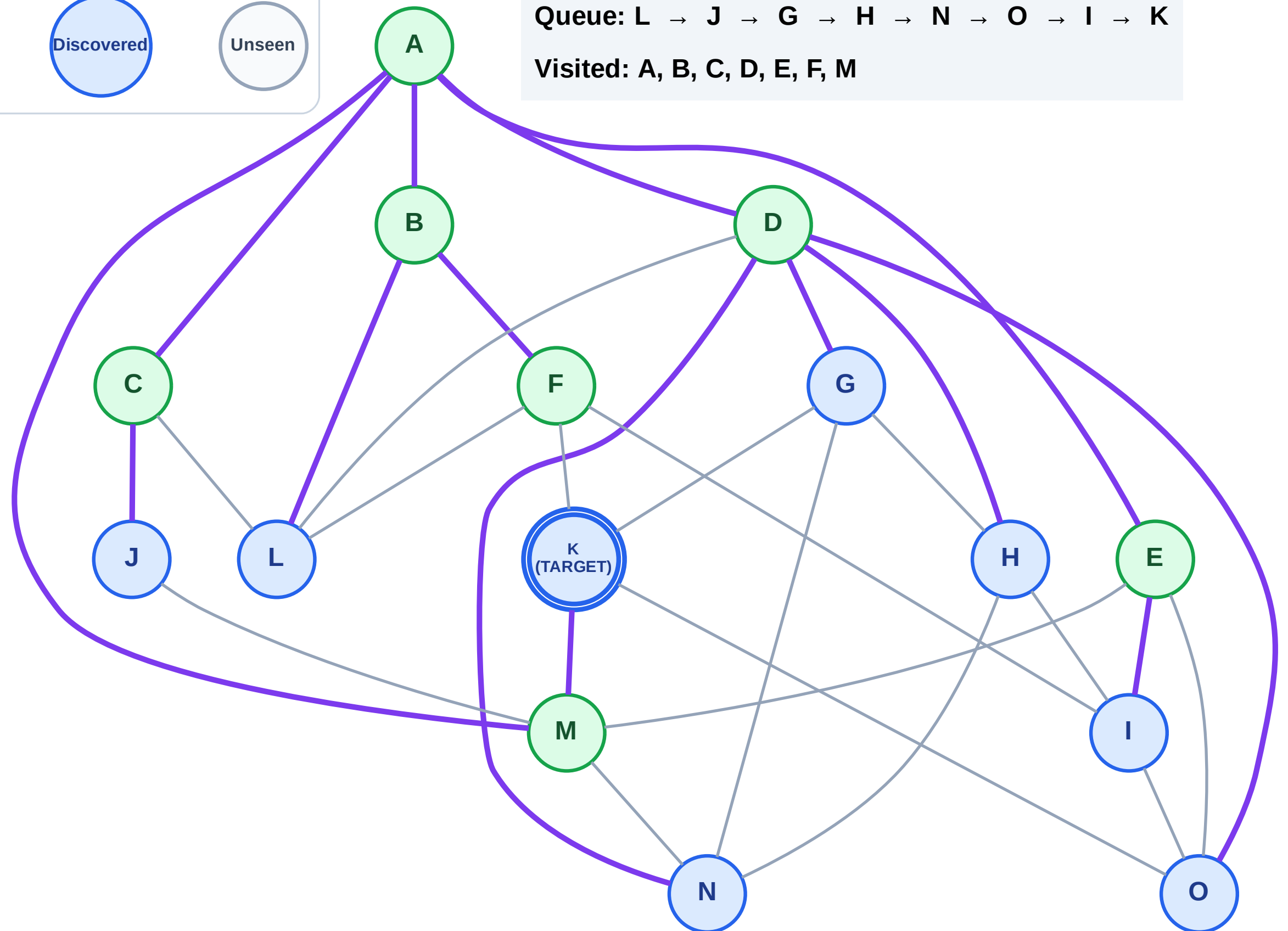
Step 15: No new neighbors from F

## Legend



Queue: L → J → G → H → N → O → I → K

Visited: A, B, C, D, E, F, M



# BFS Traversal

Step 16: Process node L

Legend

Complete

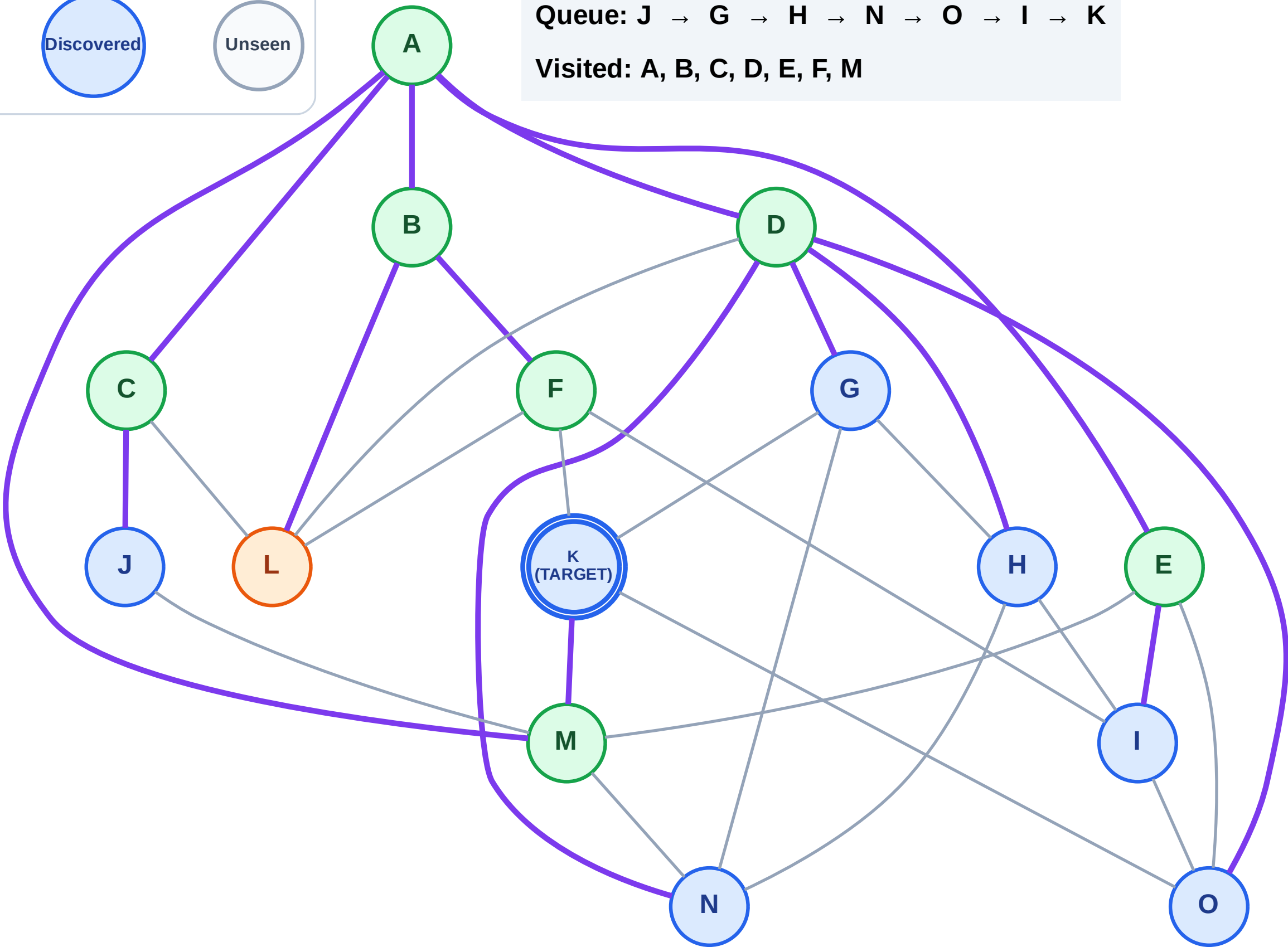
Processing

Discovered

Unseen

Queue: J → G → H → N → O → I → K

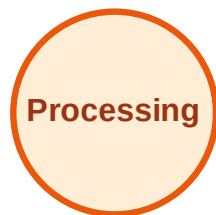
Visited: A, B, C, D, E, F, M



# BFS Traversal

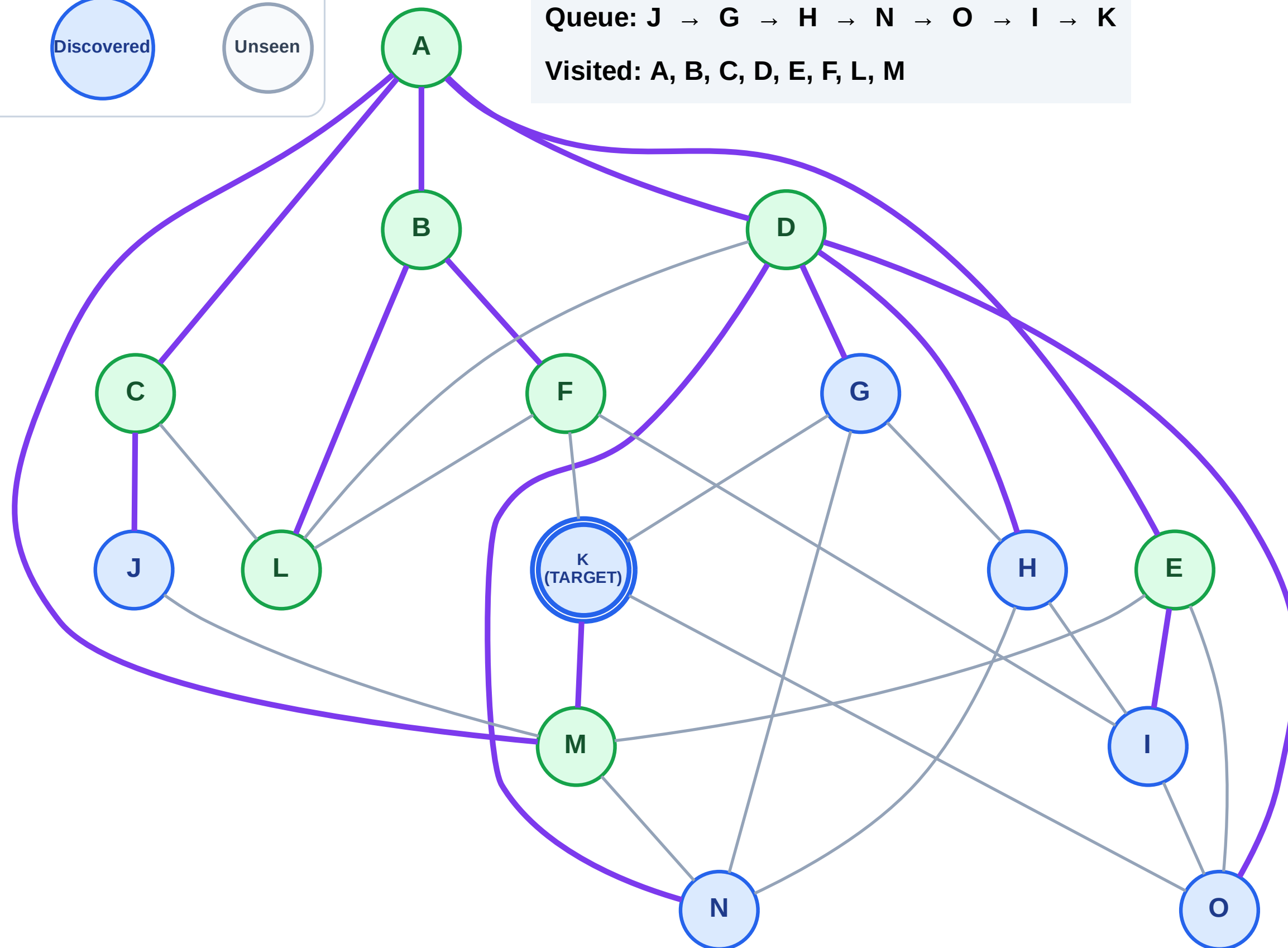
Step 17: No new neighbors from L

## Legend



Queue: J → G → H → N → O → I → K

Visited: A, B, C, D, E, F, L, M



# BFS Traversal

Step 18: Process node J

Legend

Complete

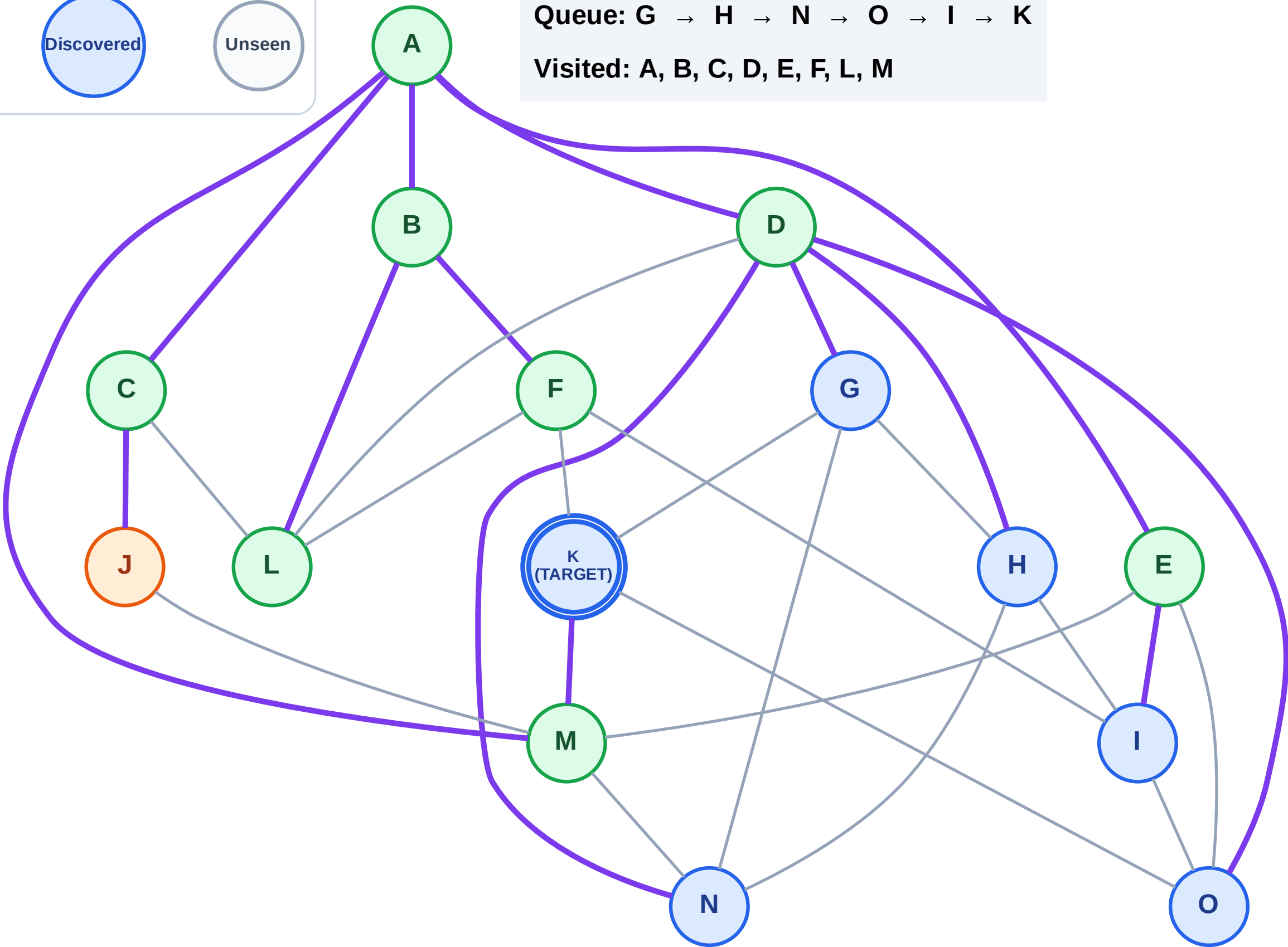
Processing

Discovered

Unseen

Queue: G → H → N → O → I → K

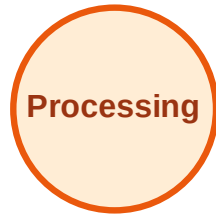
Visited: A, B, C, D, E, F, L, M



# BFS Traversal

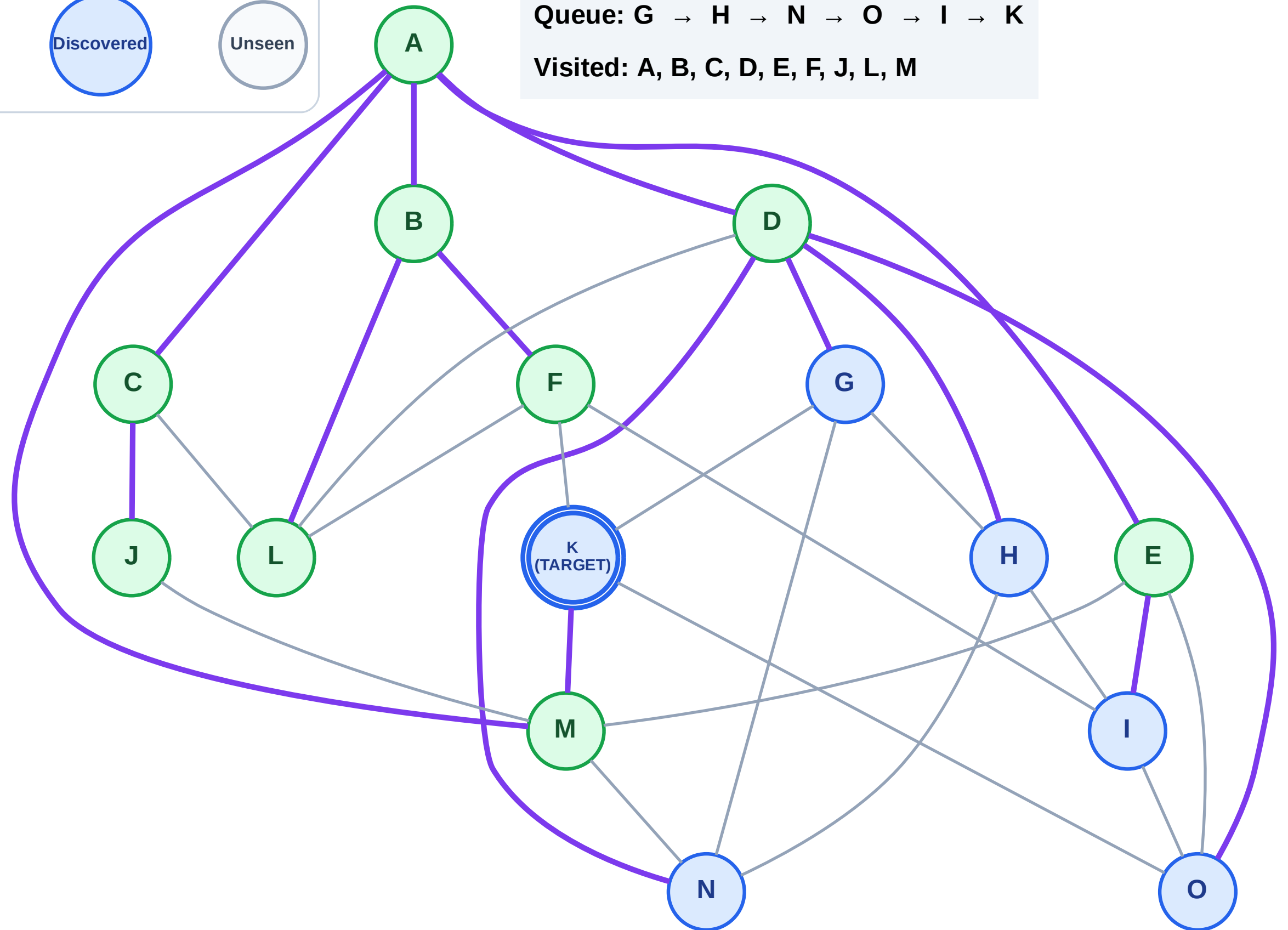
Step 19: No new neighbors from J

## Legend



Queue: G → H → N → O → I → K

Visited: A, B, C, D, E, F, J, L, M



# BFS Traversal

Step 20: Process node G

Legend

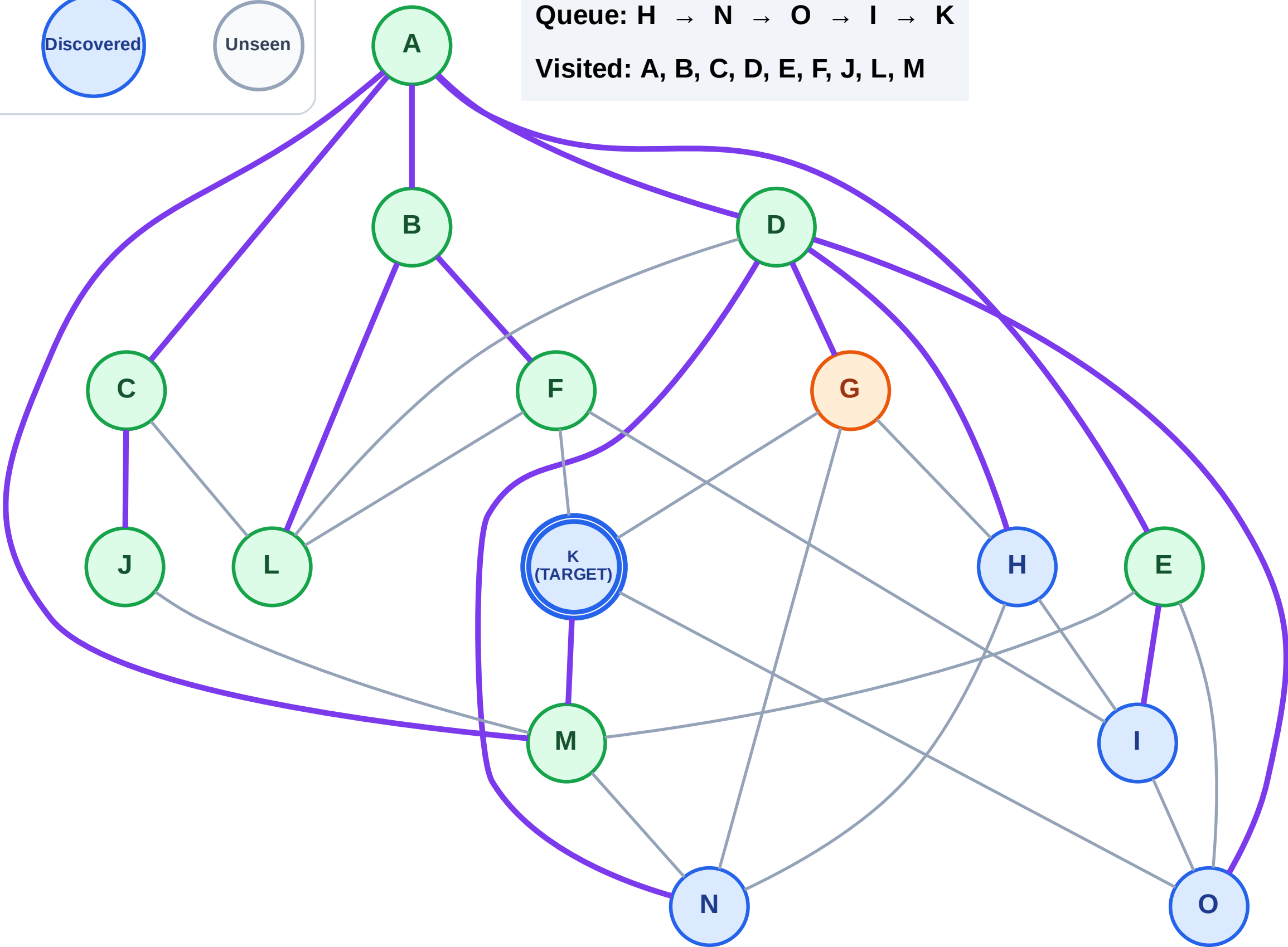
Complete

Processing

Discovered

Unseen

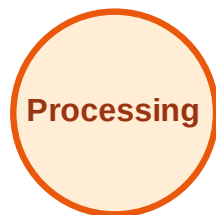
Queue: H → N → O → I → K  
Visited: A, B, C, D, E, F, J, L, M



# BFS Traversal

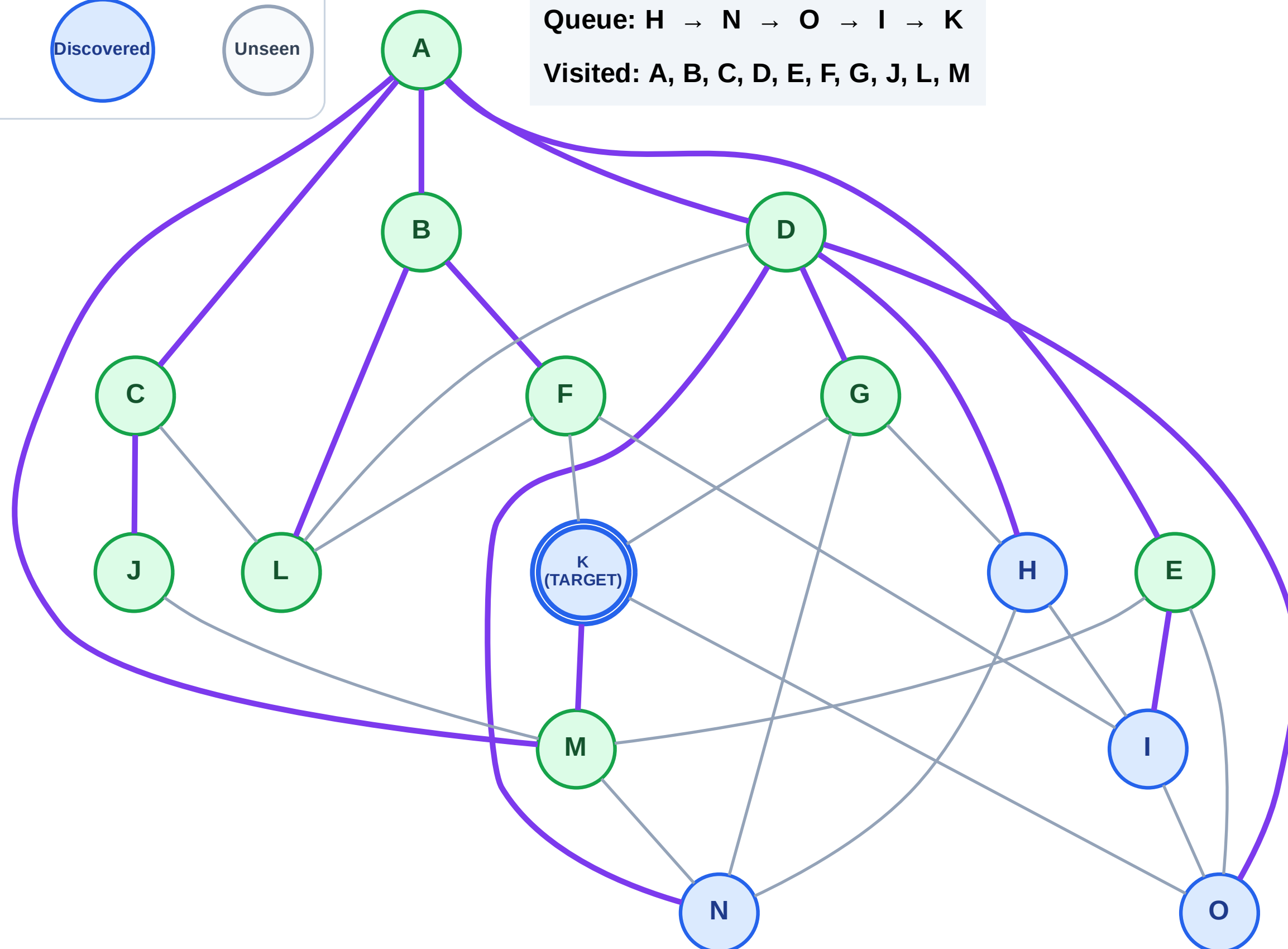
Step 21: No new neighbors from G

## Legend



Queue: H → N → O → I → K

Visited: A, B, C, D, E, F, G, J, L, M



# BFS Traversal

Step 22: Process node H

Legend

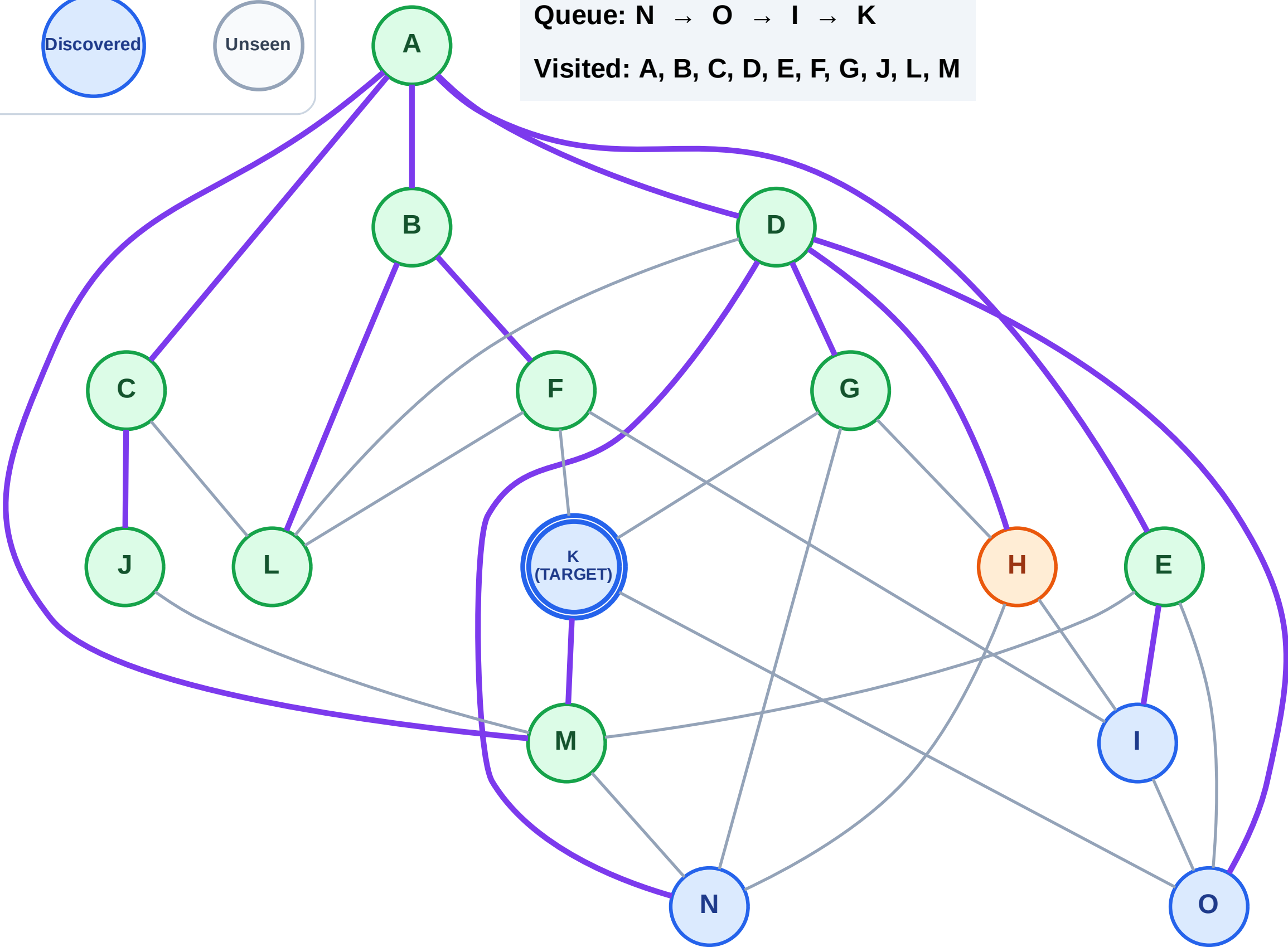
Complete

Processing

Discovered

Unseen

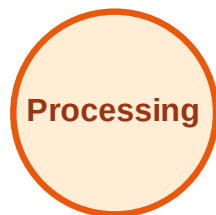
Queue: N → O → I → K  
Visited: A, B, C, D, E, F, G, J, L, M



# BFS Traversal

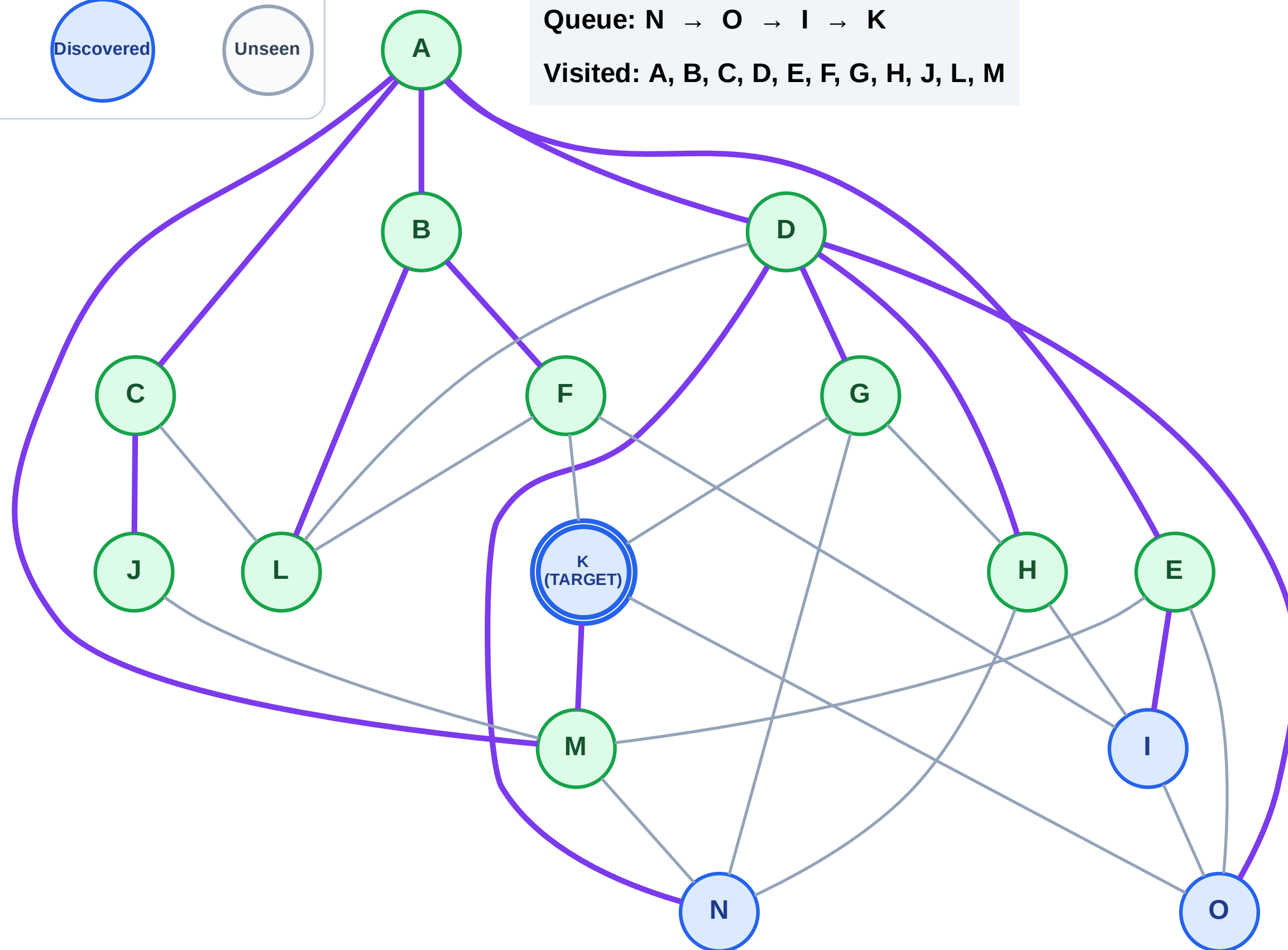
Step 23: No new neighbors from H

## Legend



Queue: N → O → I → K

Visited: A, B, C, D, E, F, G, H, J, L, M



# BFS Traversal

Step 24: Process node N

Legend

Complete

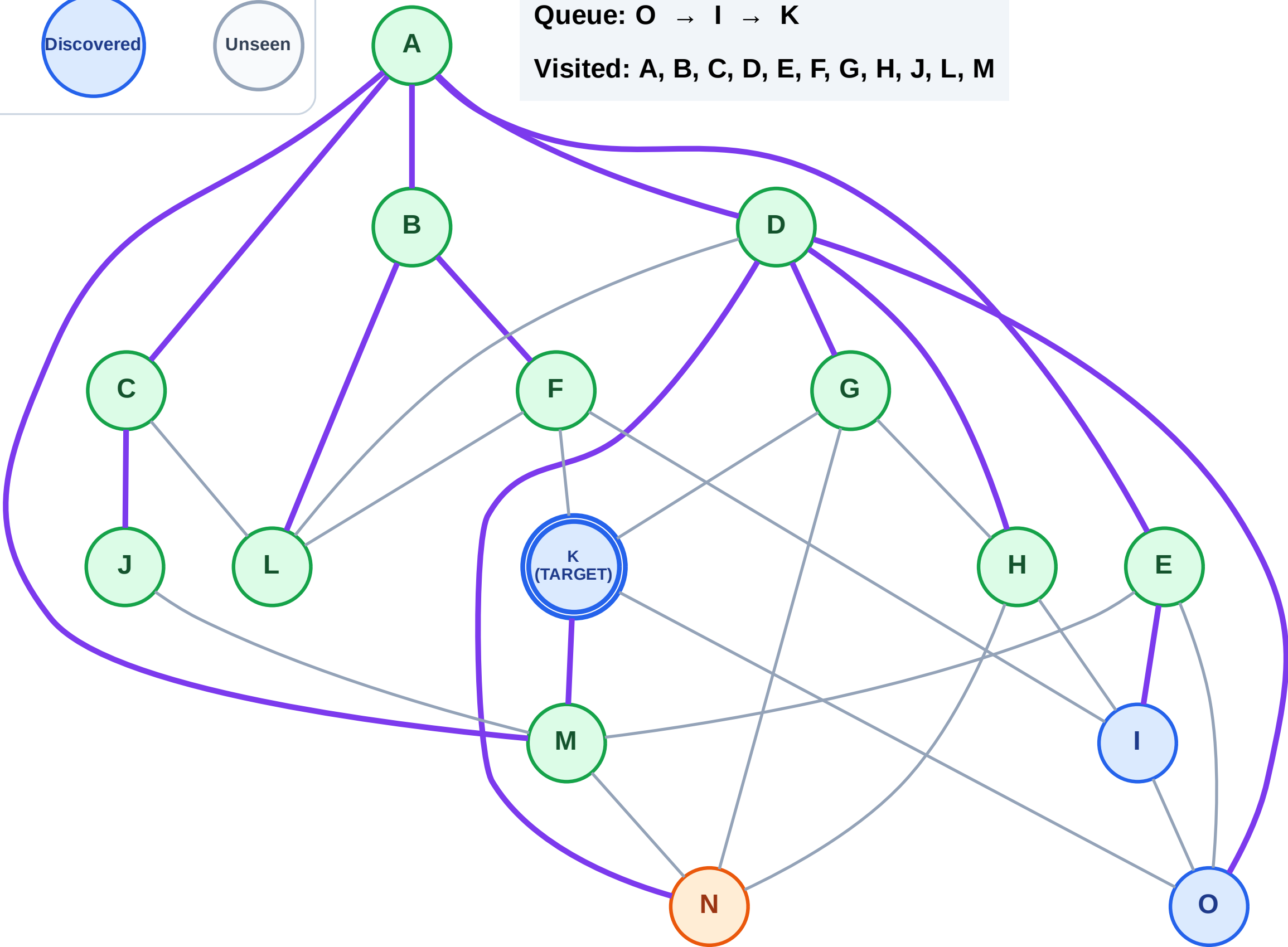
Processing

Discovered

Unseen

Queue: O → I → K

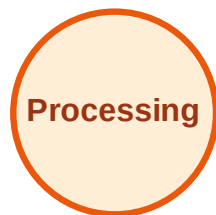
Visited: A, B, C, D, E, F, G, H, J, L, M



# BFS Traversal

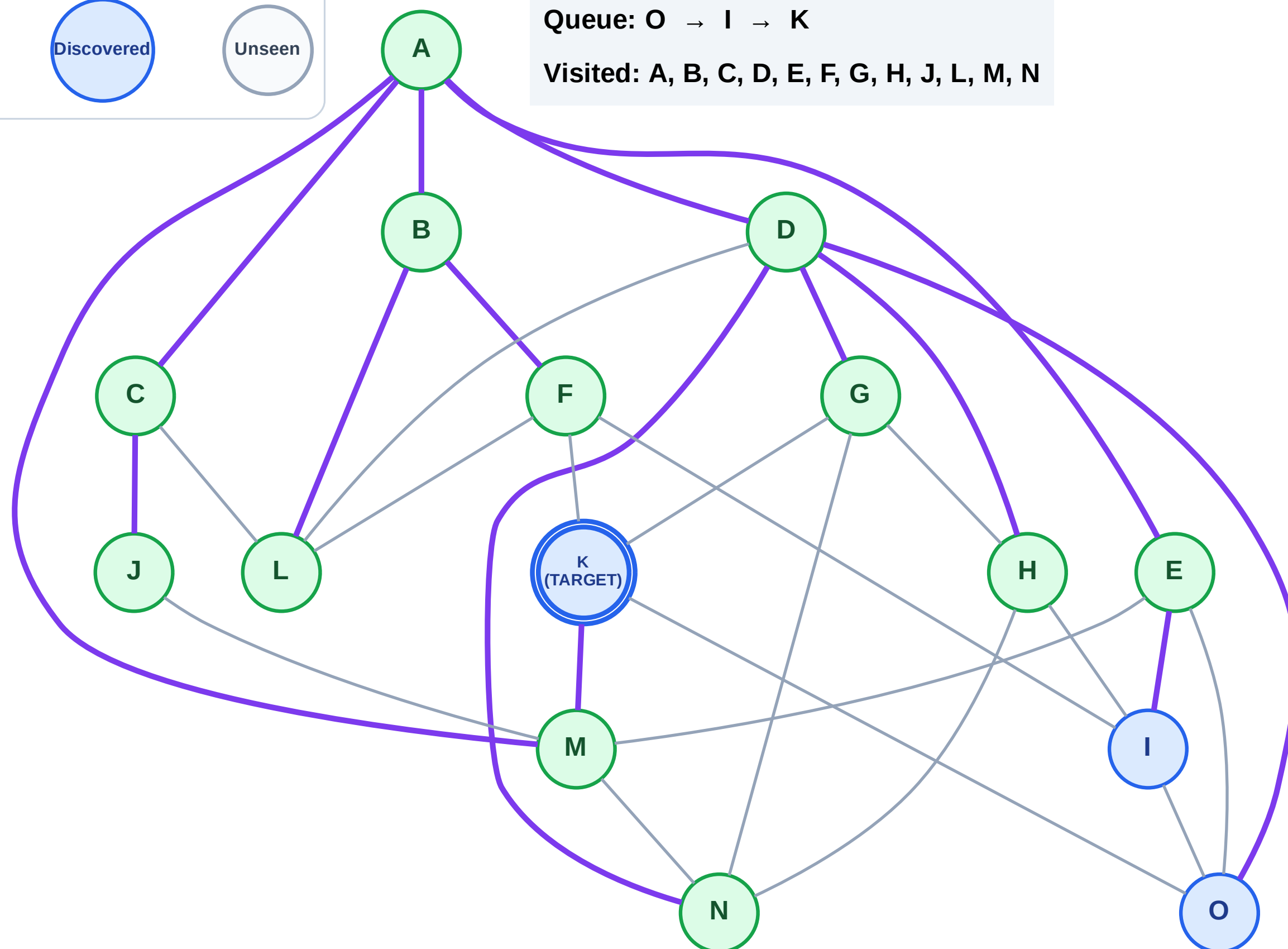
Step 25: No new neighbors from N

## Legend



Queue: O → I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N



# BFS Traversal

Step 26: Process node O

Legend

Complete

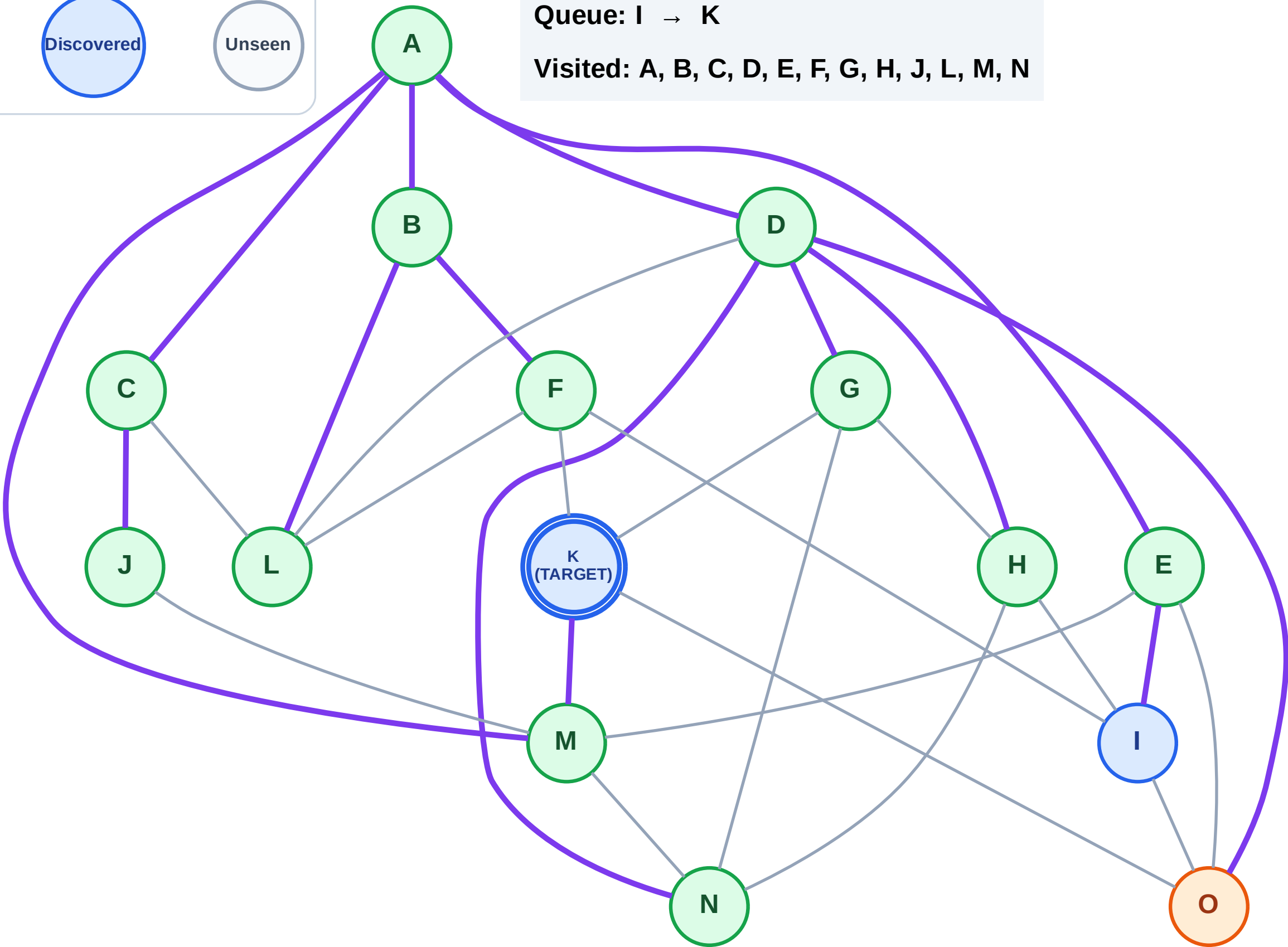
Processing

Discovered

Unseen

Queue: I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N



# BFS Traversal

Step 27: No new neighbors from O

Legend

Complete

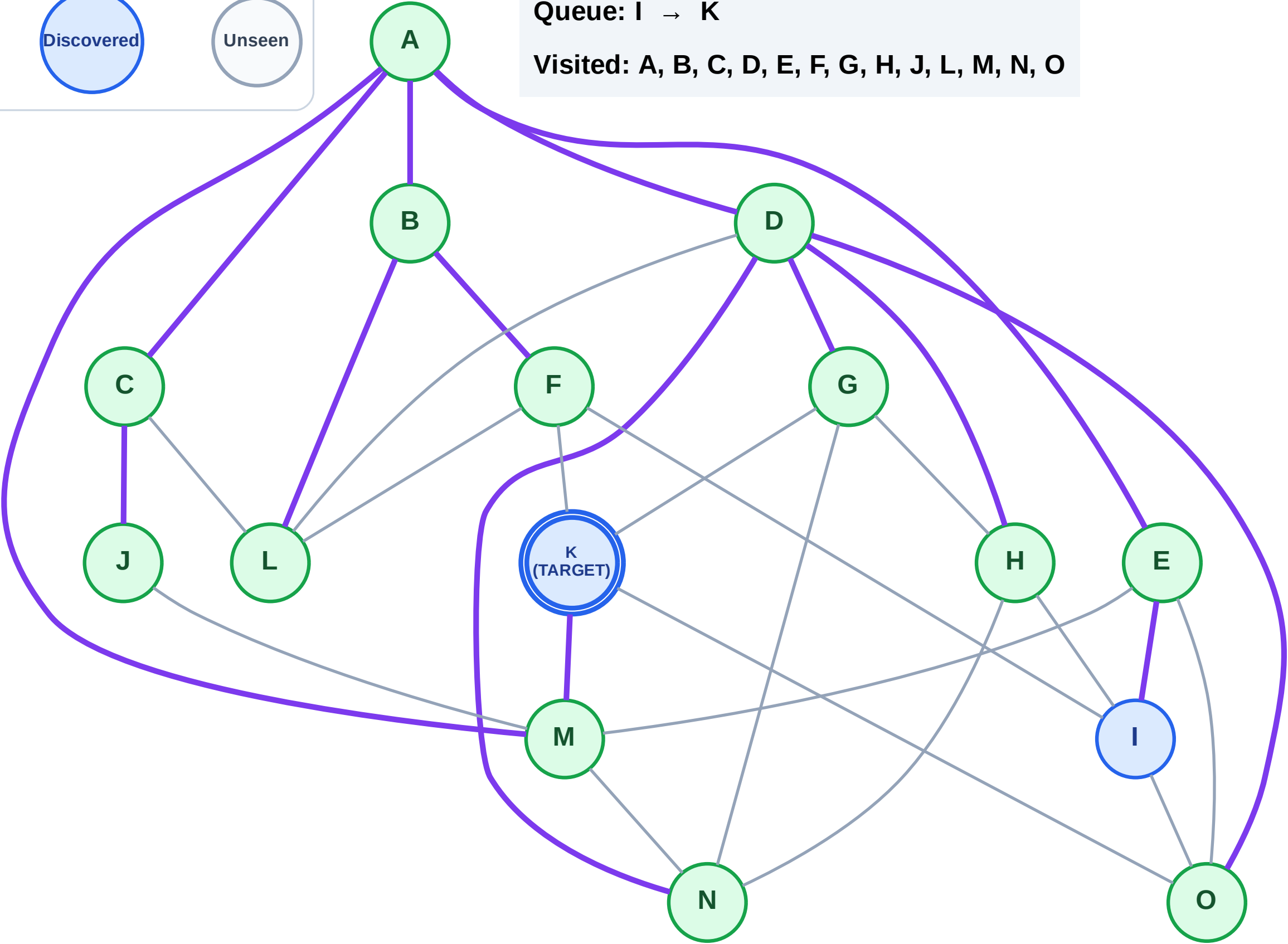
Processing

Discovered

Unseen

Queue: I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



# BFS Traversal

Step 28: Process node I

Legend

Complete

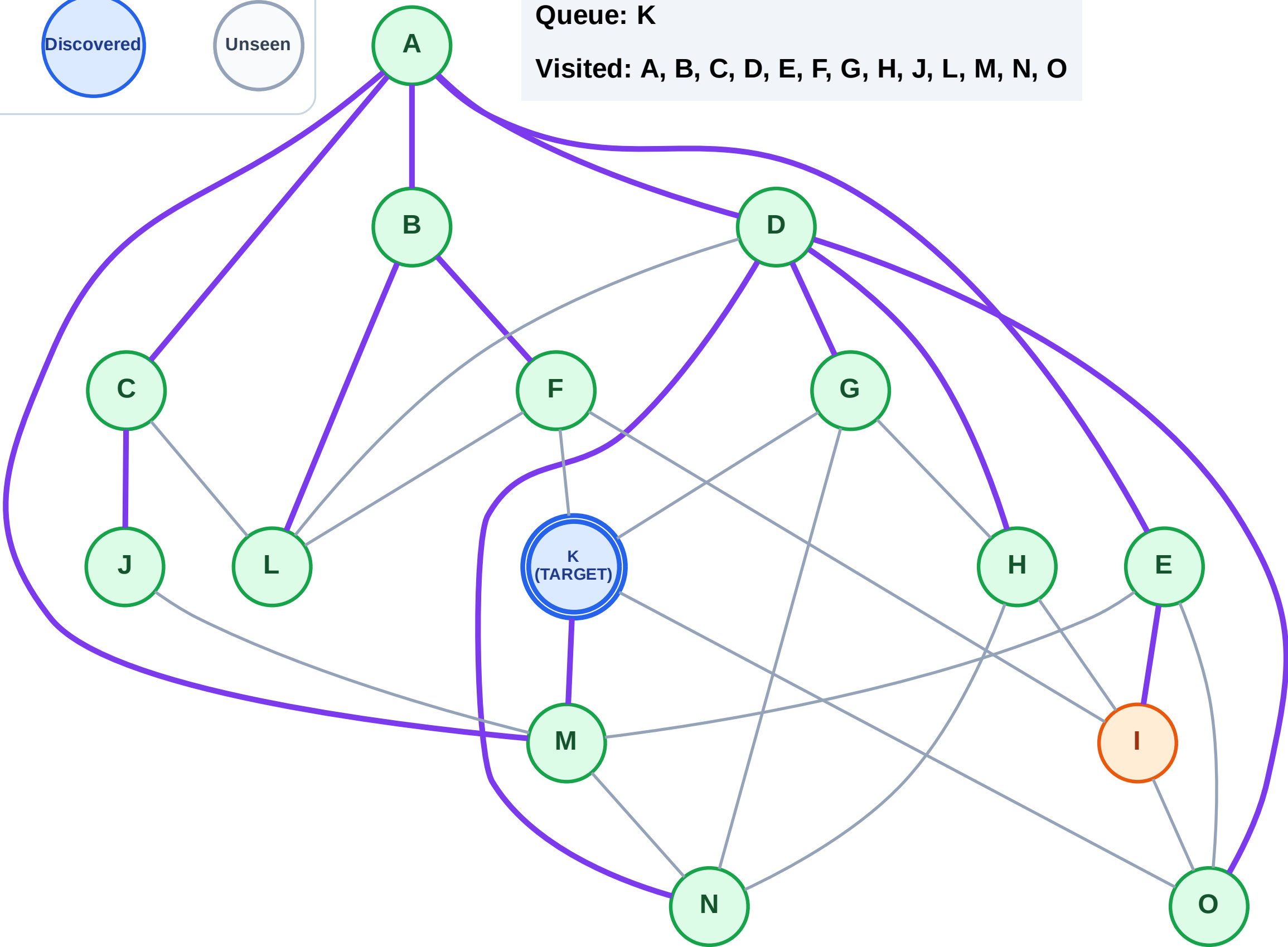
Processing

Discovered

Unseen

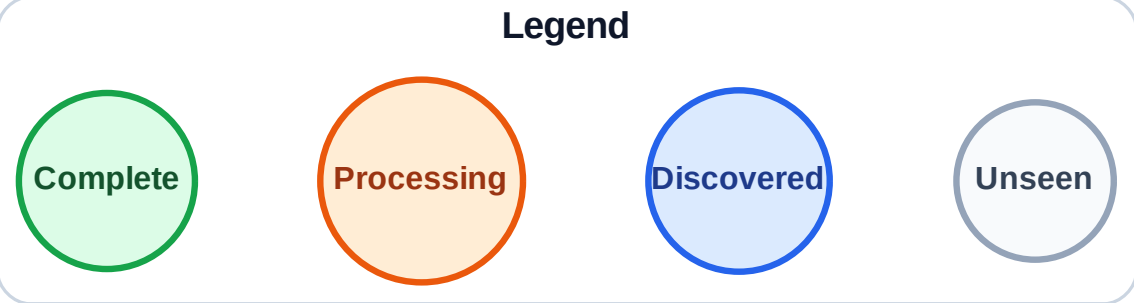
Queue: K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



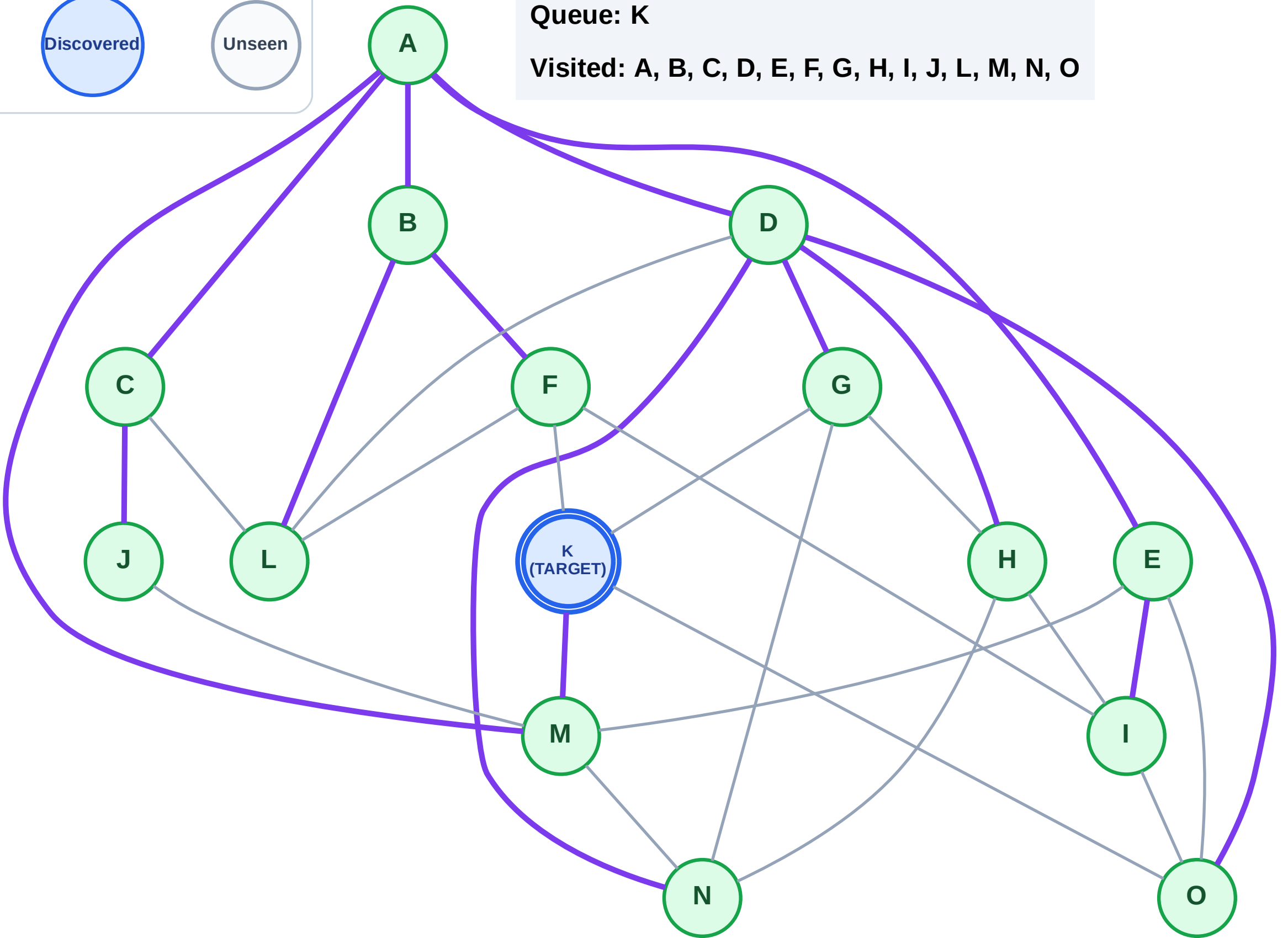
# BFS Traversal

## Step 29: No new neighbors from I



## Queue: K

**Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O**



# BFS Traversal

Step 30: Process node K

Legend

Complete

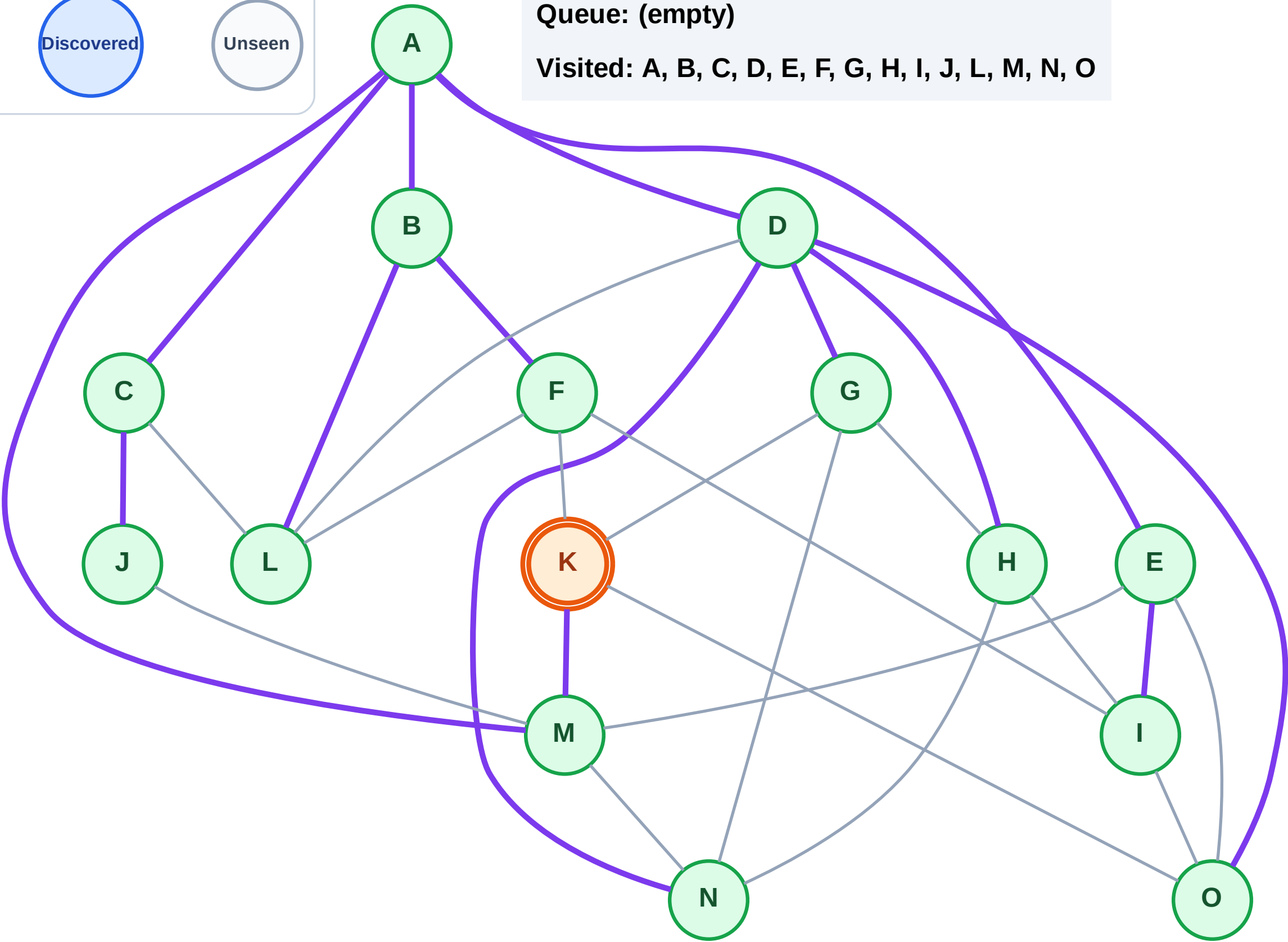
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



# BFS Traversal

Step 31: TARGET NODE K REACHED! Path (2 hops): A → M → K

Legend

Complete

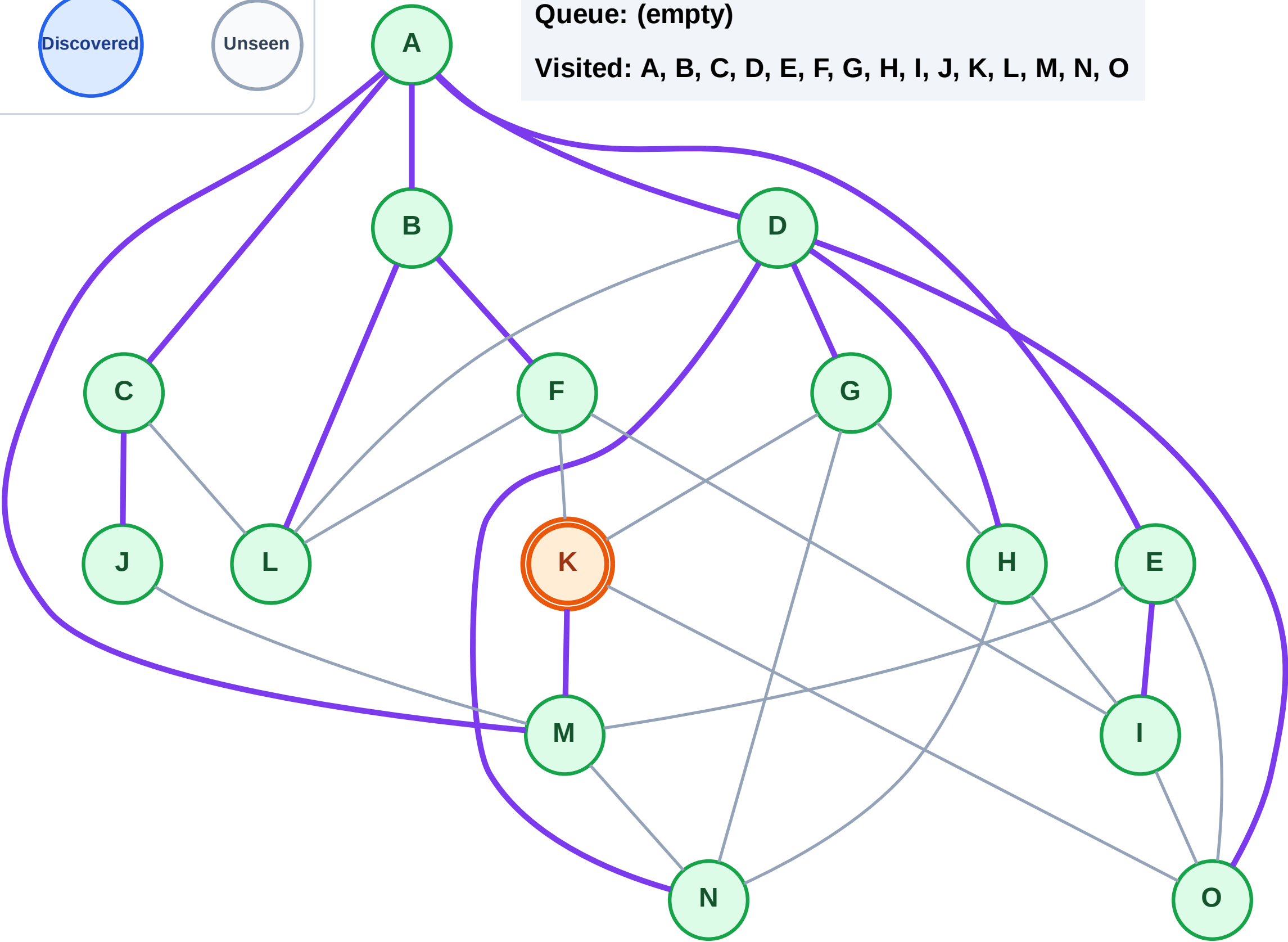
Processing

Discovered

Unseen

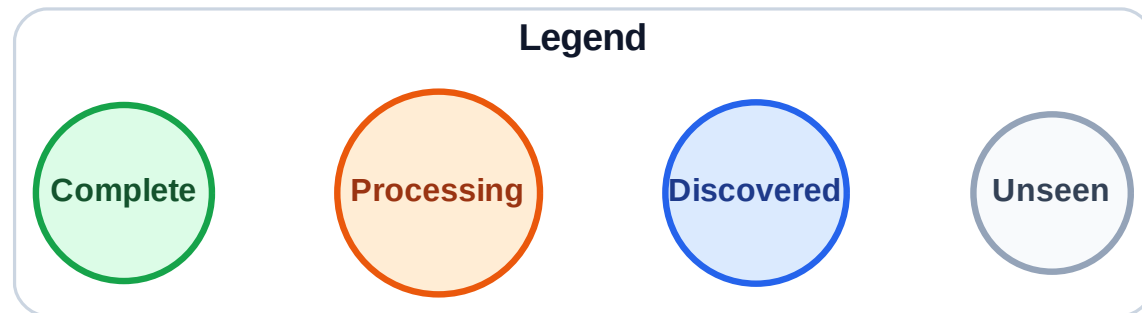
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

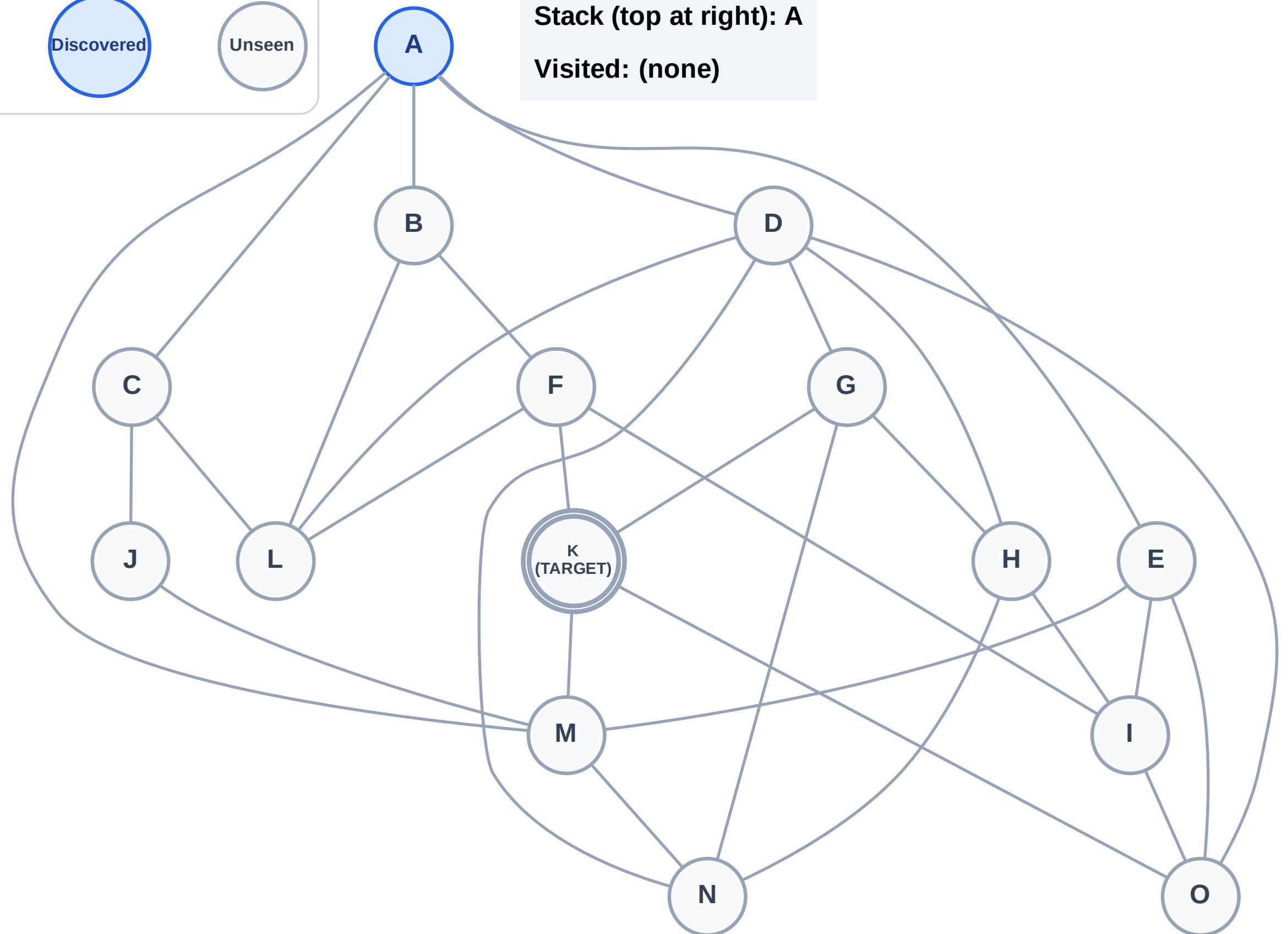


# DFS Traversal

Step 1: Start at A (Seeking target node K)



**Stack (top at right): A**  
**Visited: (none)**



# DFS Traversal

Step 2: Process node A

Legend

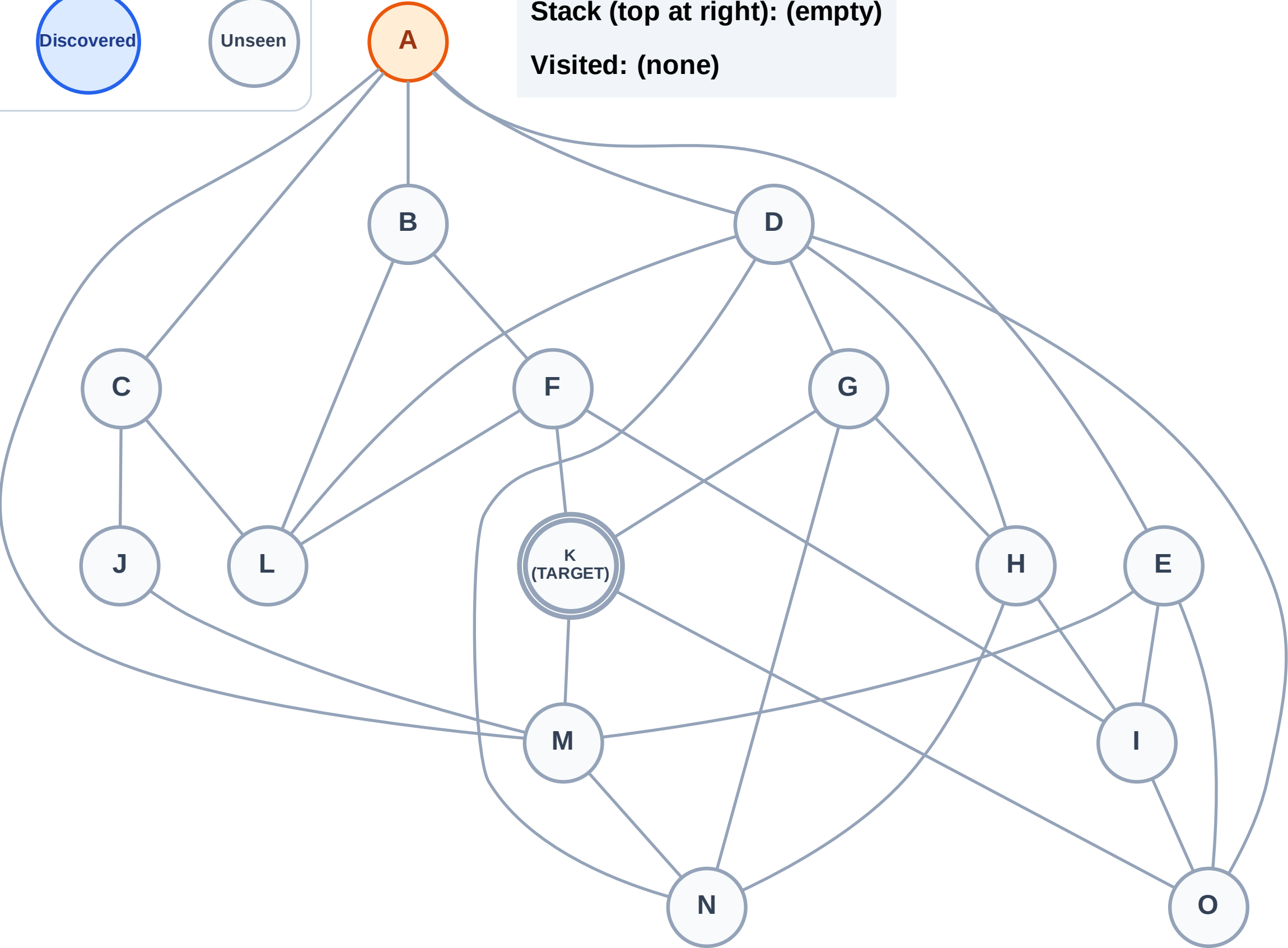
Complete

Processing

Discovered

Unseen

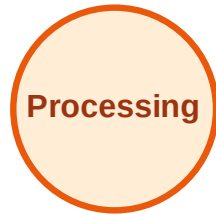
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

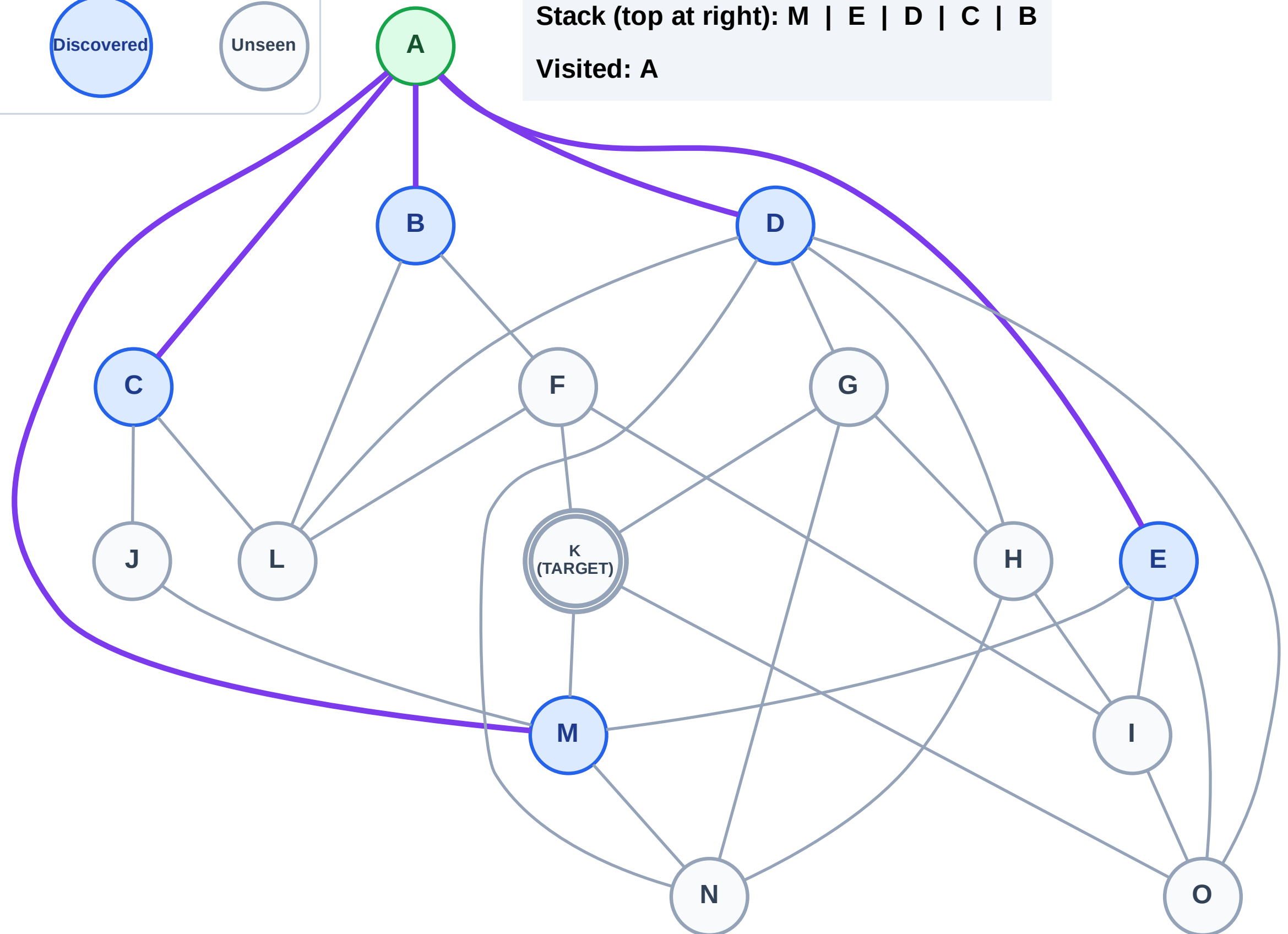
Step 3: Discover M, E, D, C, B from A

## Legend



Stack (top at right): M | E | D | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

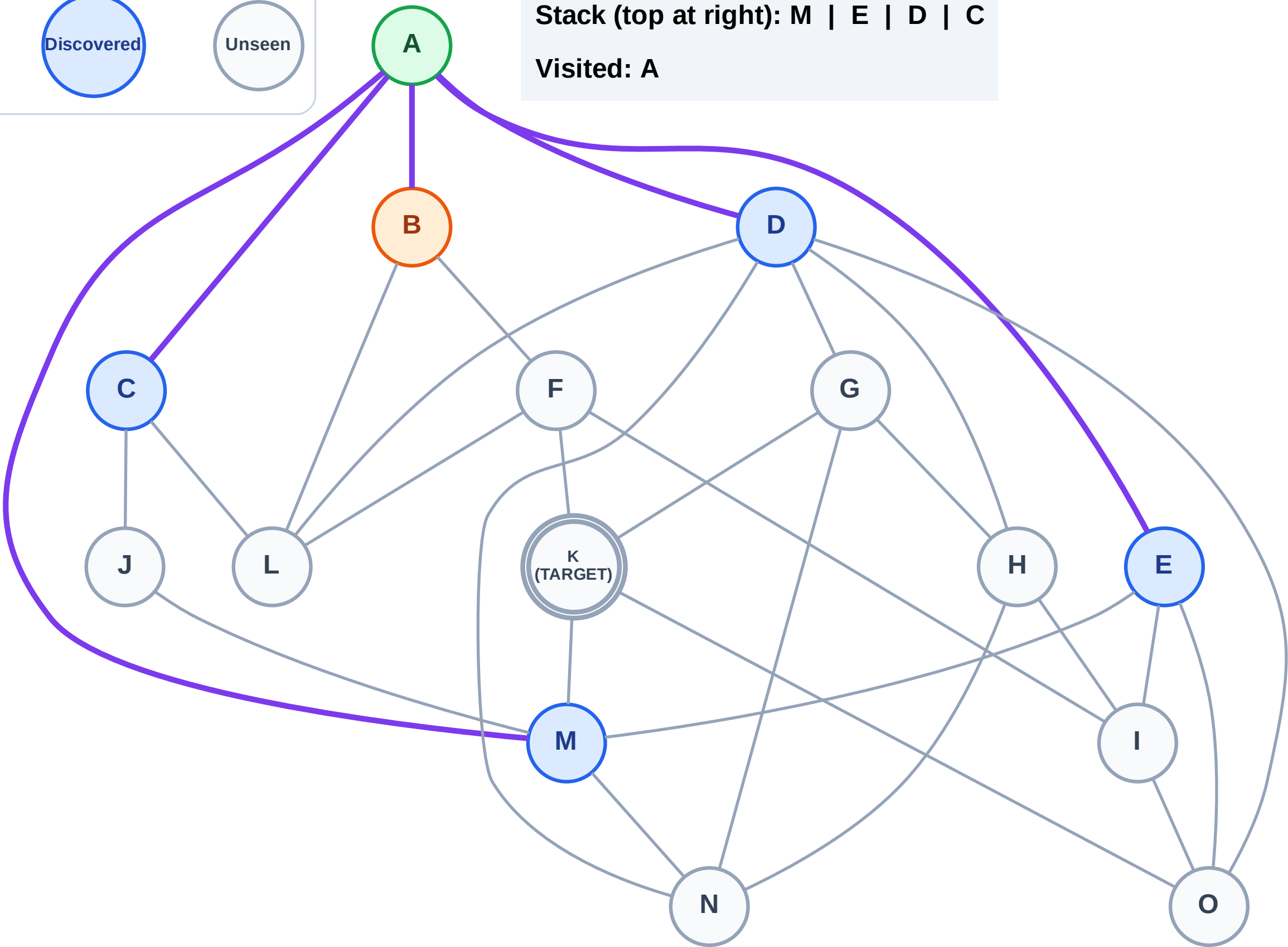
Complete

Processing

Discovered

Unseen

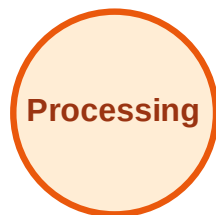
Stack (top at right): M | E | D | C  
Visited: A



# DFS Traversal

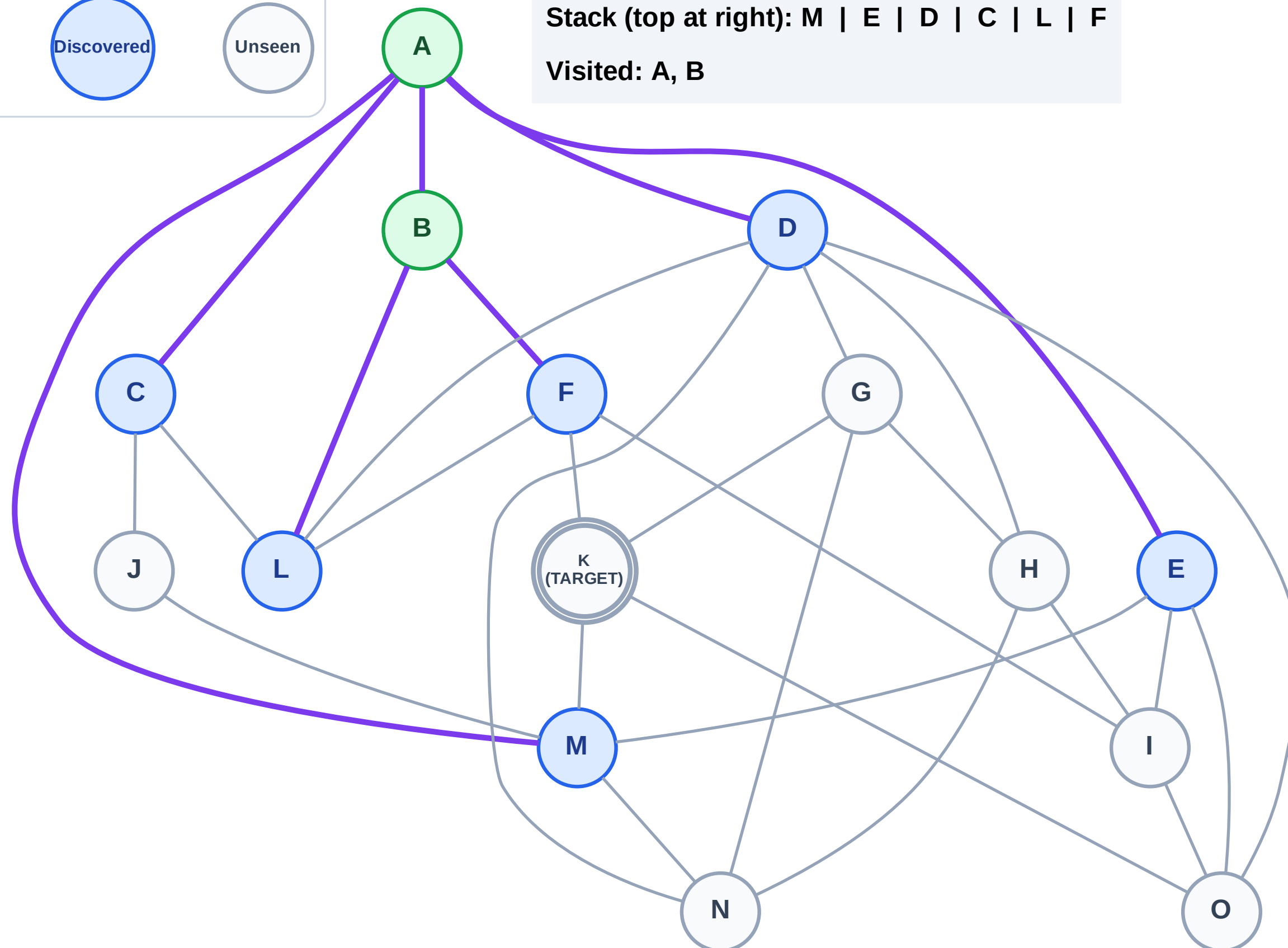
Step 5: Discover L, F from B

## Legend



Stack (top at right): M | E | D | C | L | F

Visited: A, B



# DFS Traversal

Step 6: Process node F

Legend

Complete

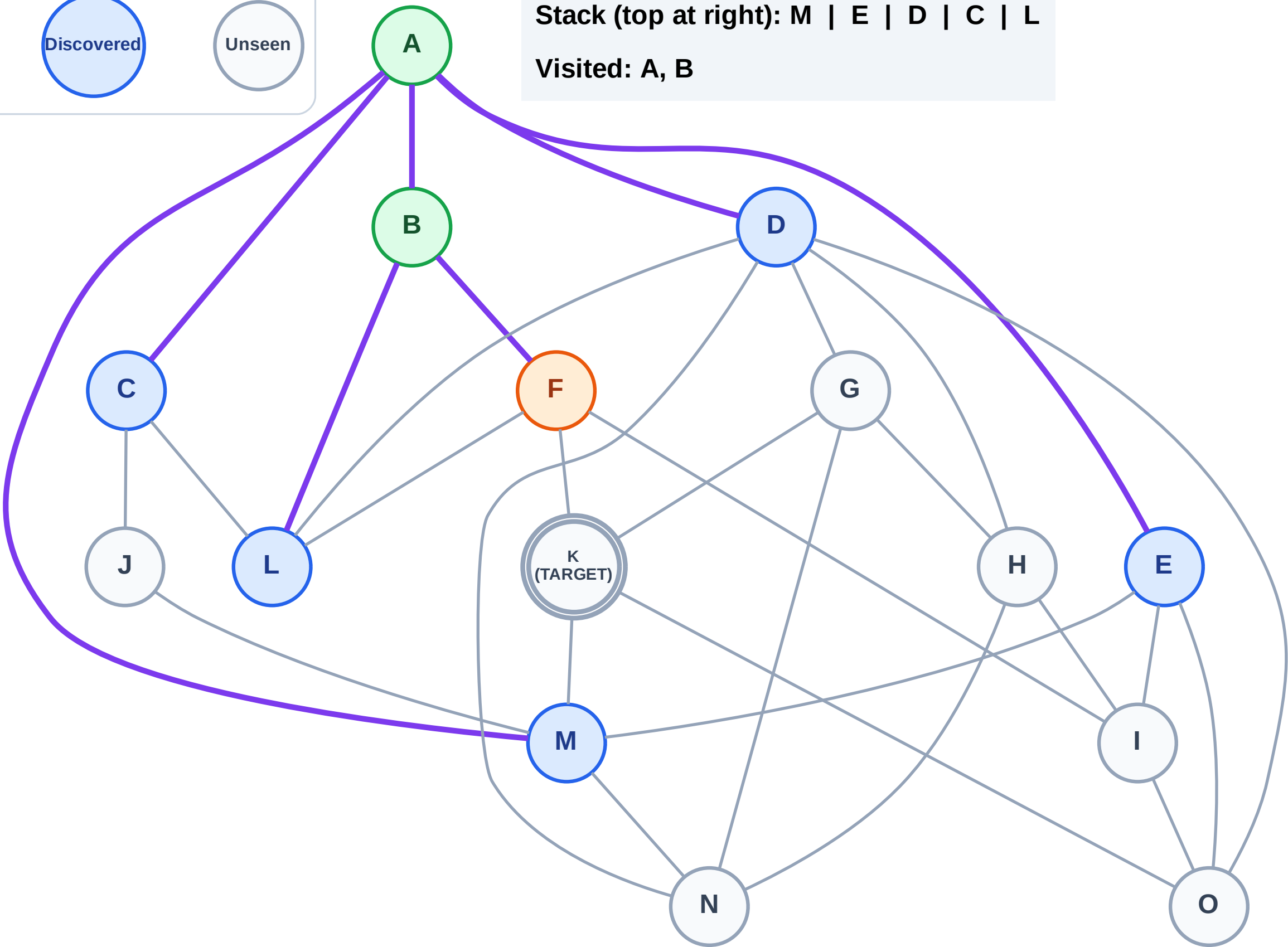
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L

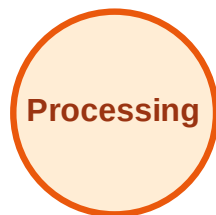
Visited: A, B



# DFS Traversal

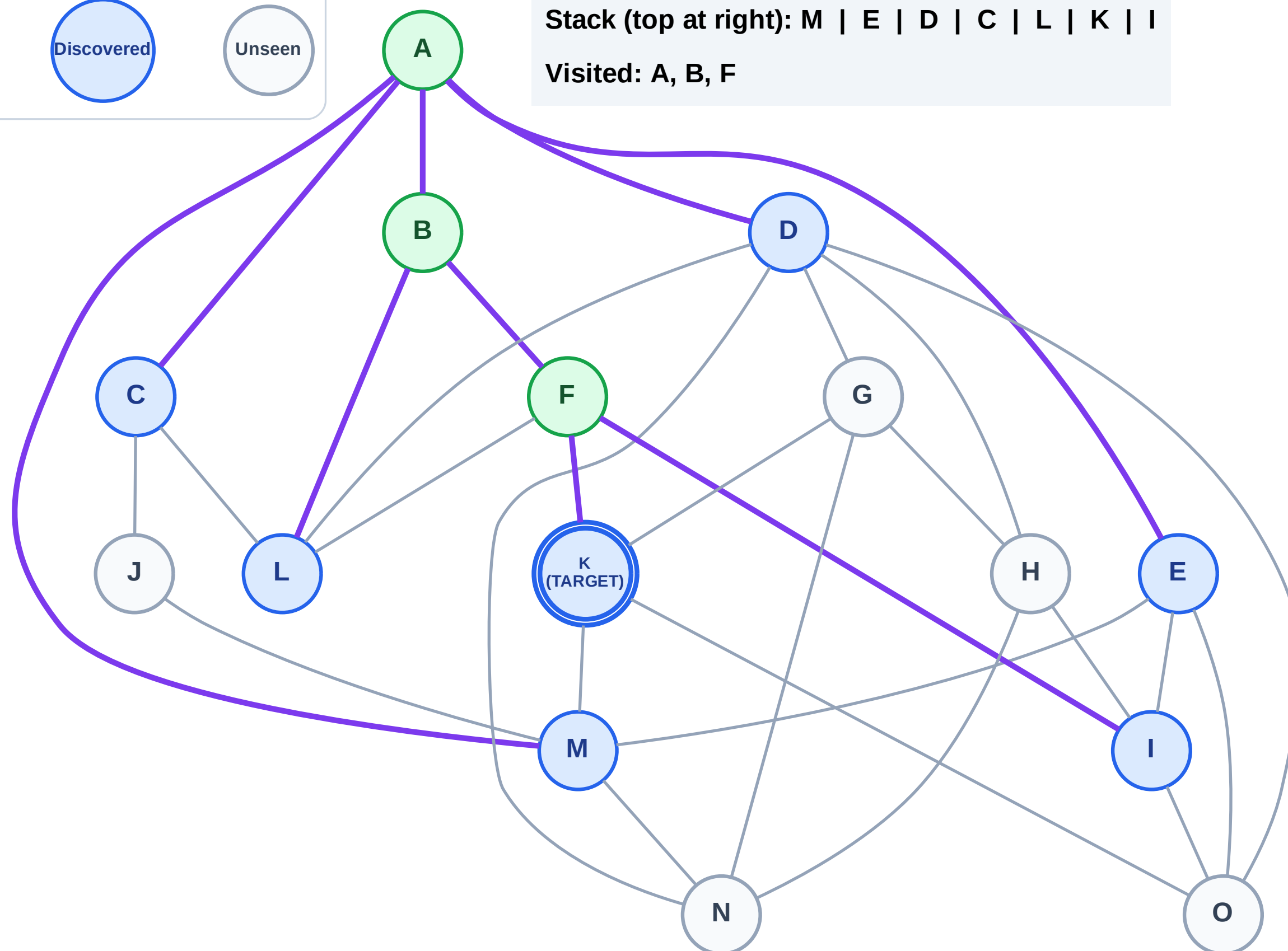
Step 7: Discover K, I from F

## Legend



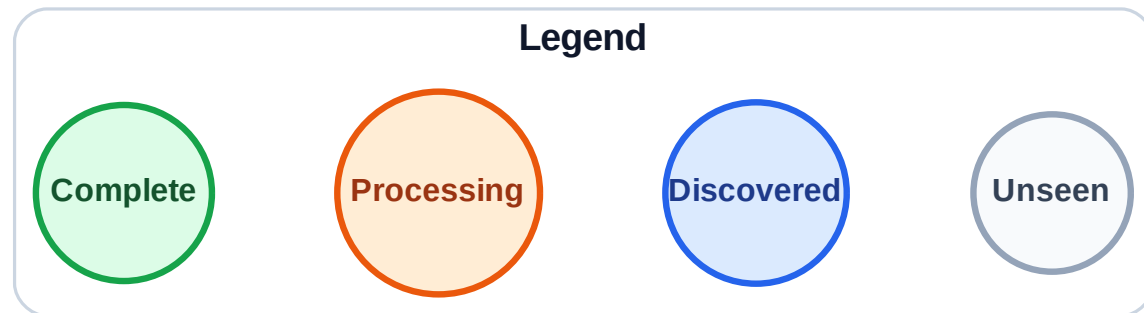
Stack (top at right): M | E | D | C | L | K | I

Visited: A, B, F



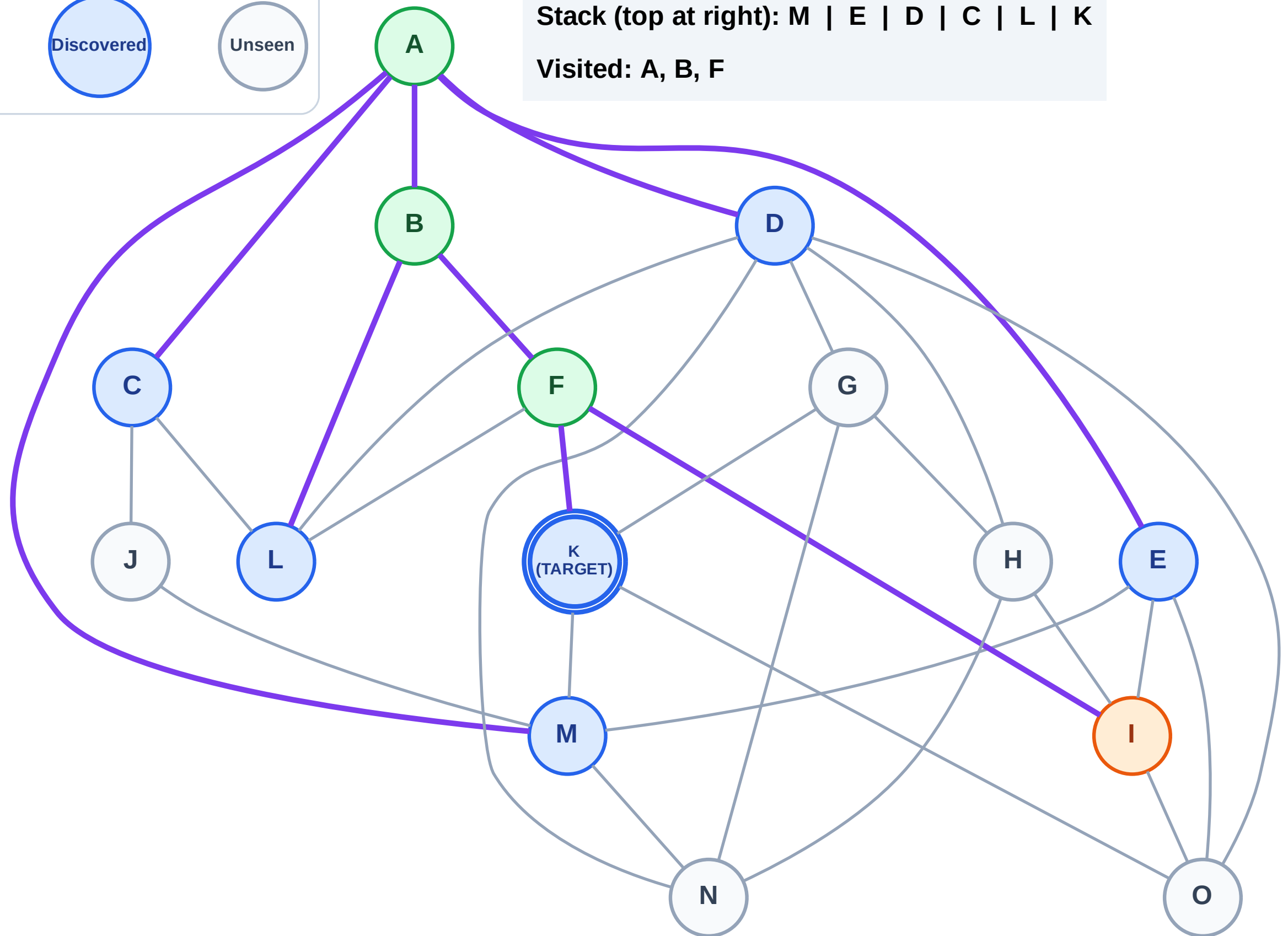
### Step 8: Process node I

### Step 8: Process node I



**Stack (top at right):** M | E | D | C | L | K

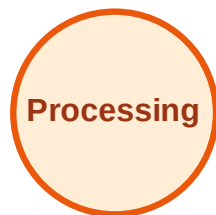
**Visited: A, B, F**



# DFS Traversal

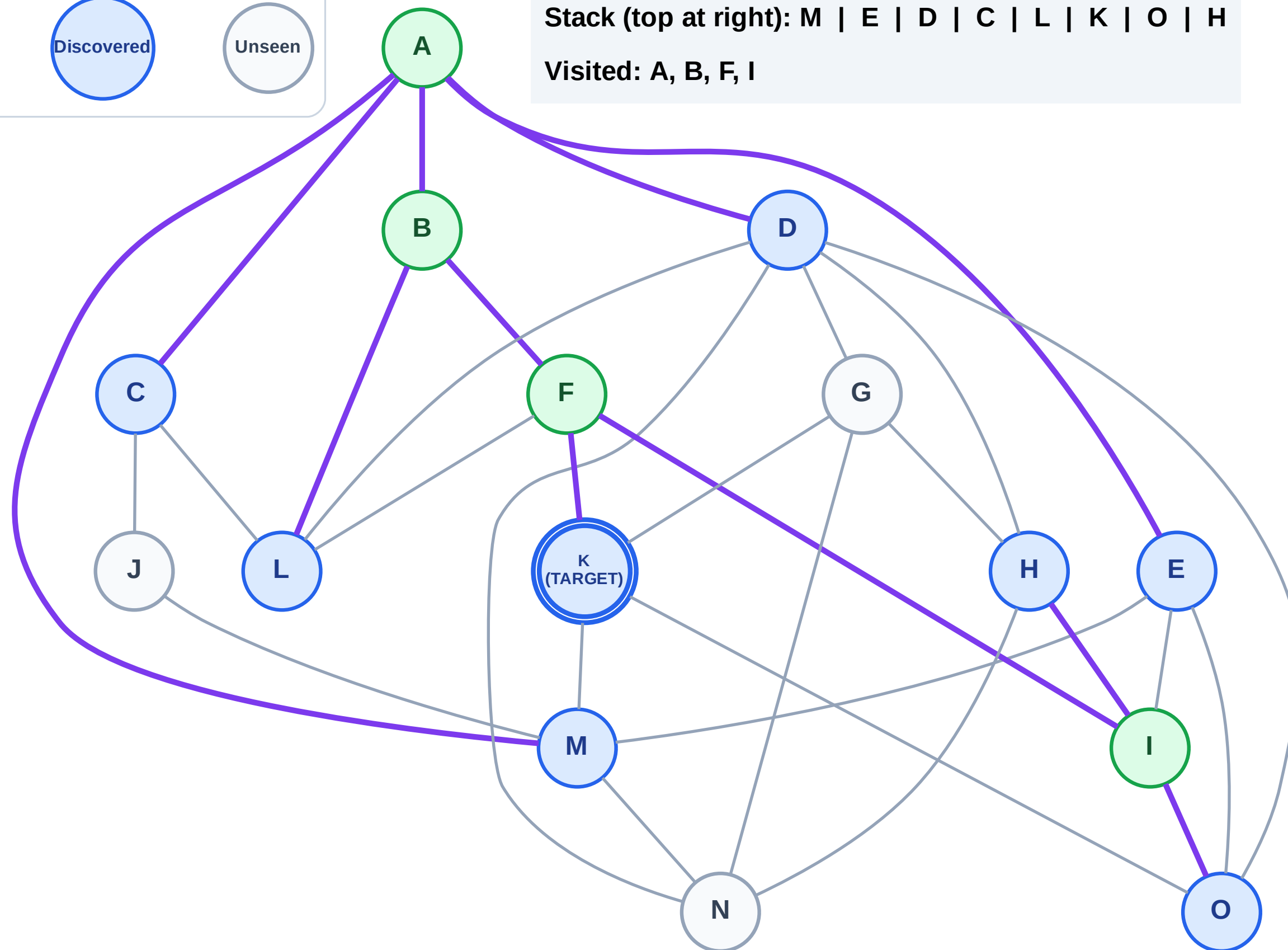
Step 9: Discover O, H from I

## Legend



Stack (top at right): M | E | D | C | L | K | O | H

Visited: A, B, F, I



# DFS Traversal

Step 10: Process node H

Legend

Complete

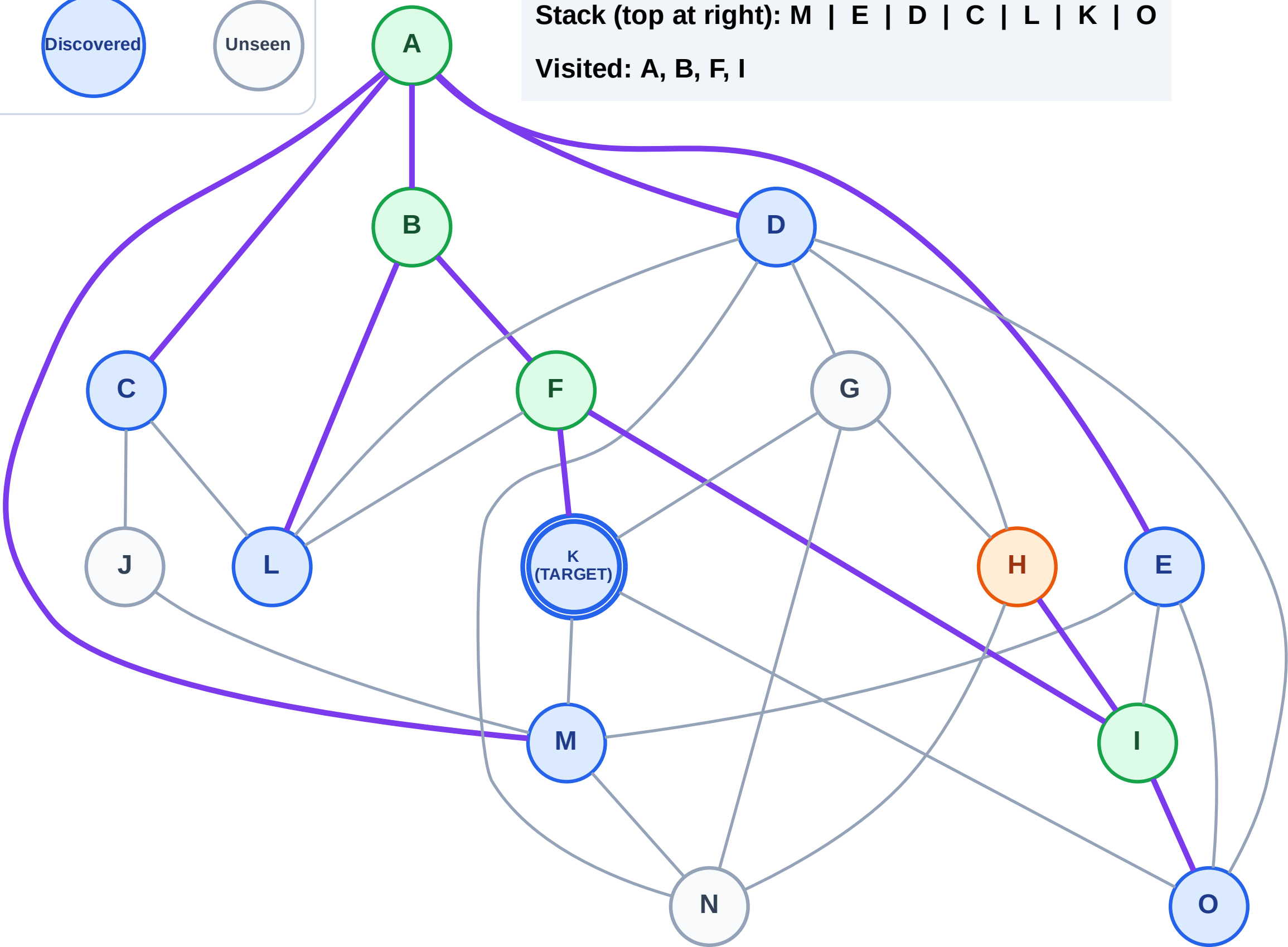
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O

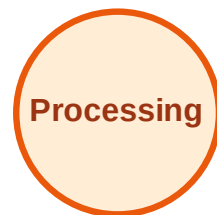
Visited: A, B, F, I



# DFS Traversal

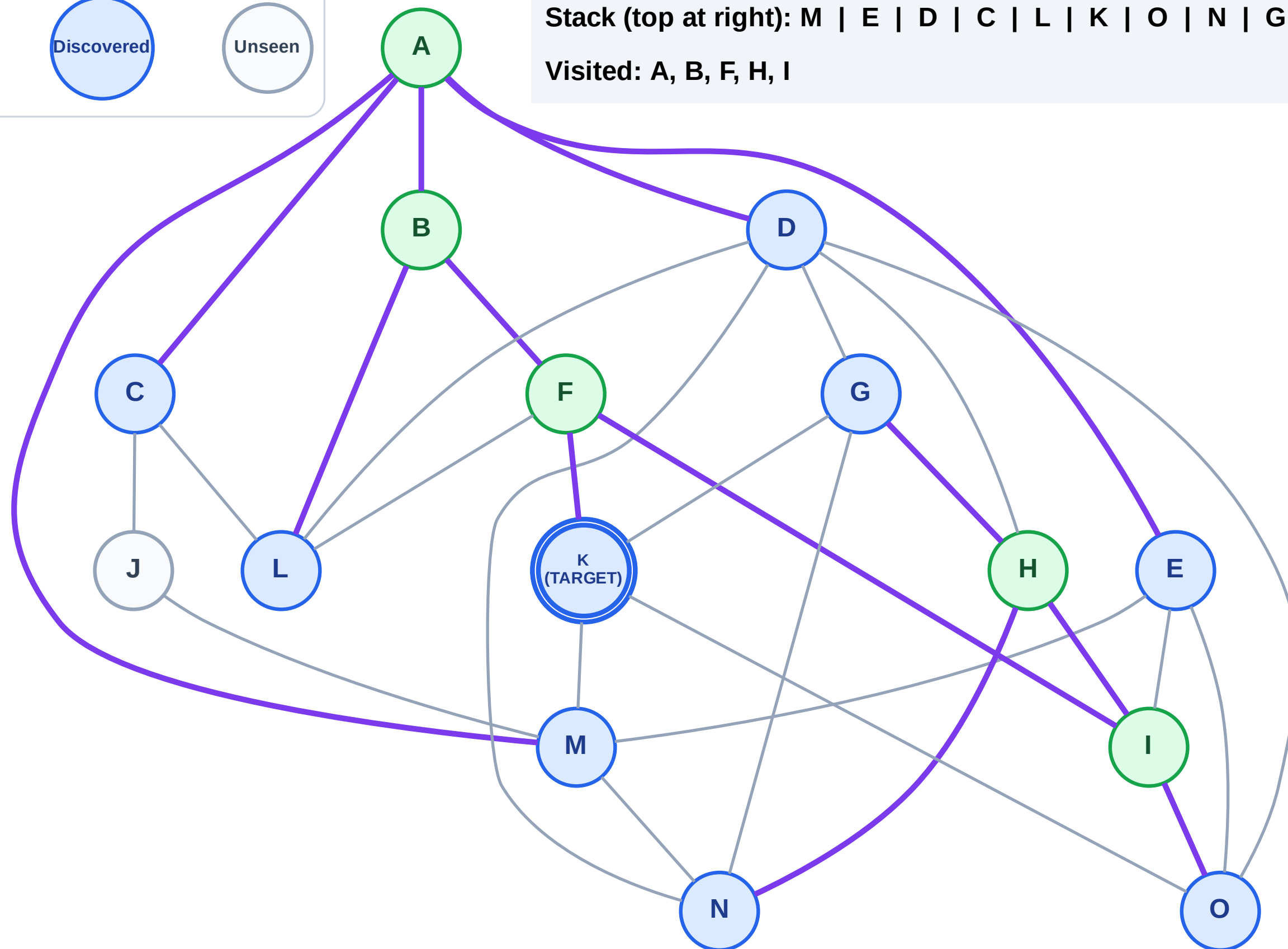
Step 11: Discover N, G from H

## Legend



Stack (top at right): M | E | D | C | L | K | O | N | G

Visited: A, B, F, H, I



# DFS Traversal

Step 12: Process node G

## Legend

Complete

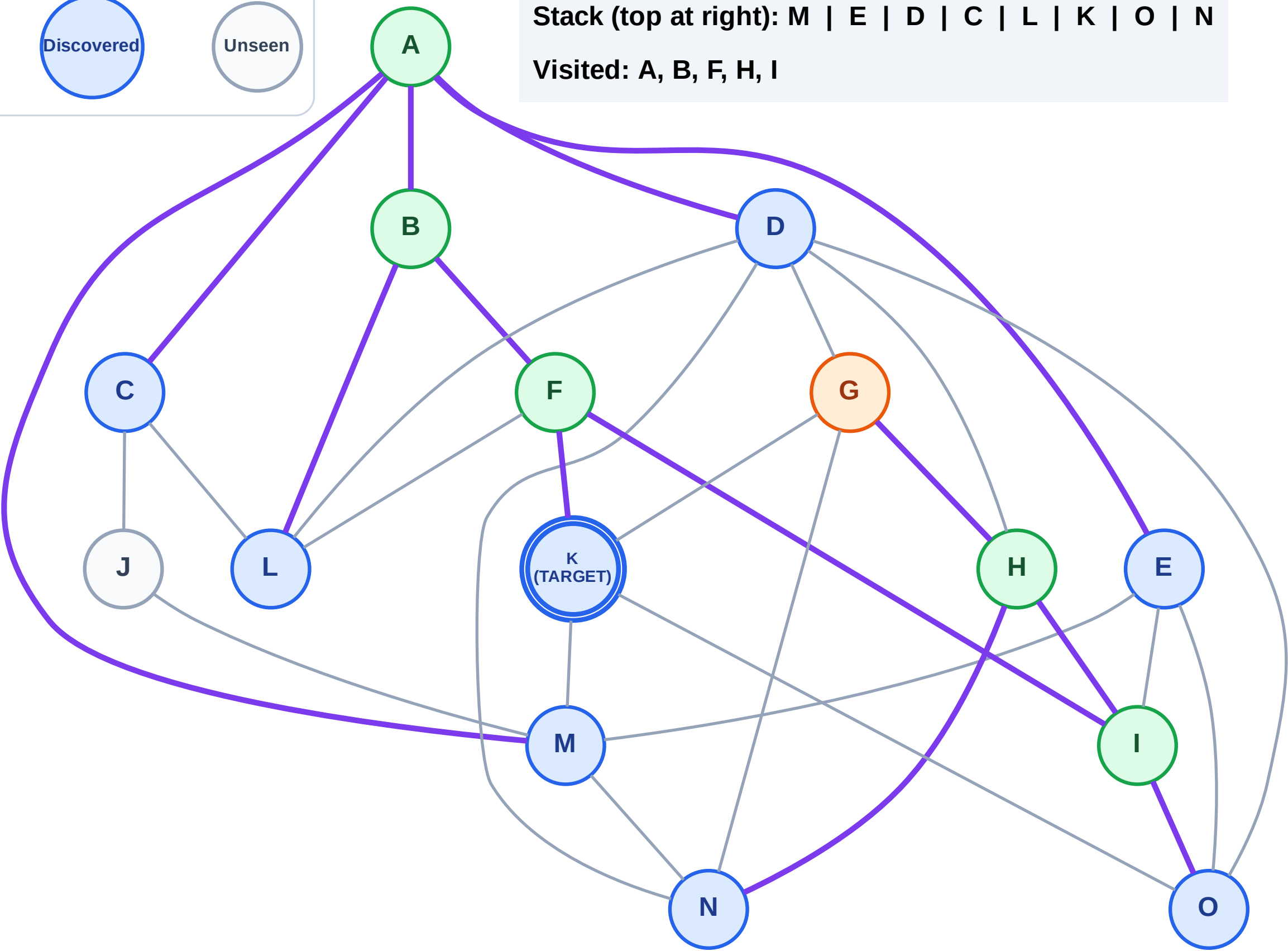
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O | N

Visited: A, B, F, H, I



# DFS Traversal

Step 13: No new neighbors from G

Legend

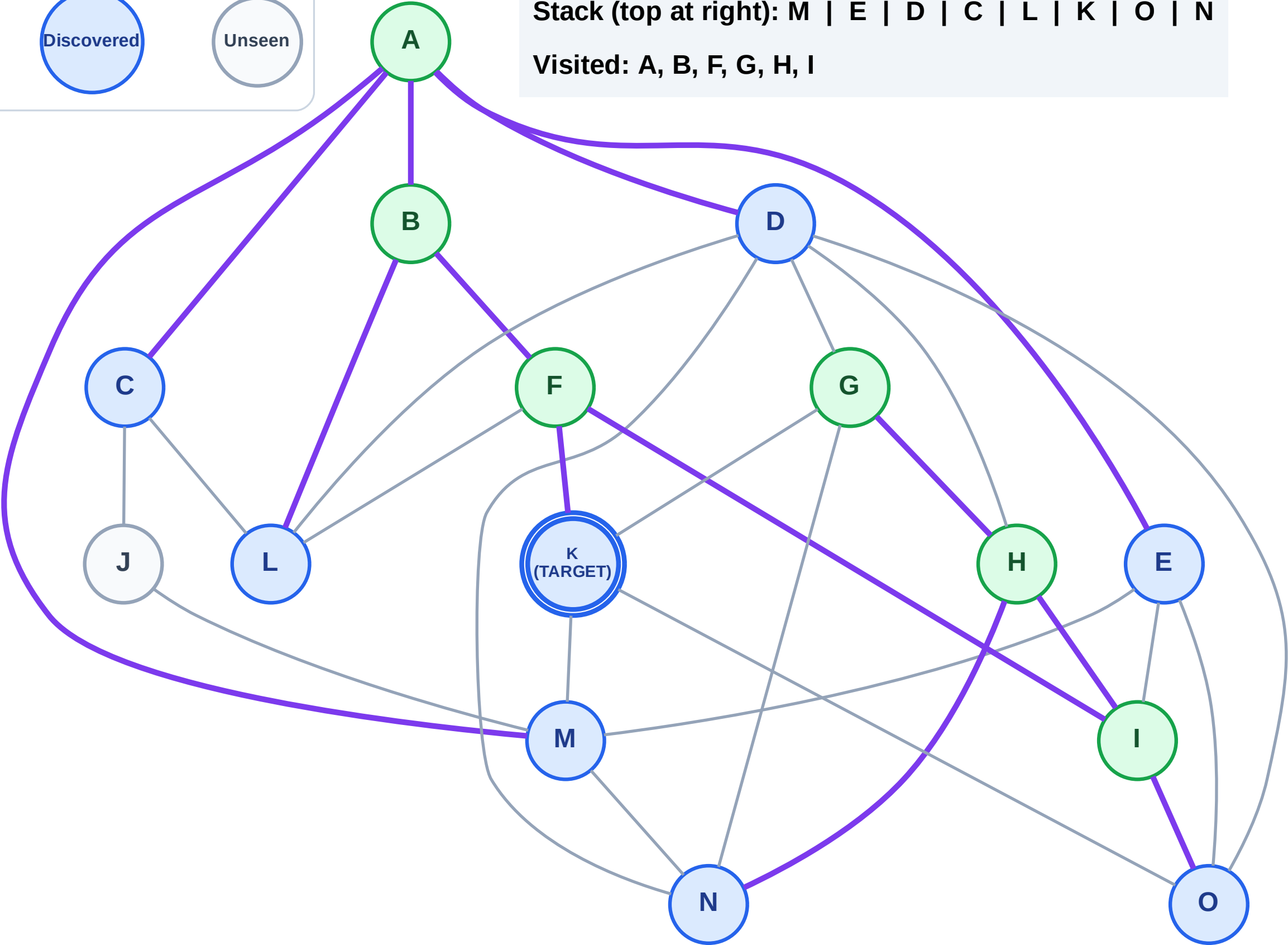
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O | N  
Visited: A, B, F, G, H, I



# DFS Traversal

Step 14: Process node N

Legend

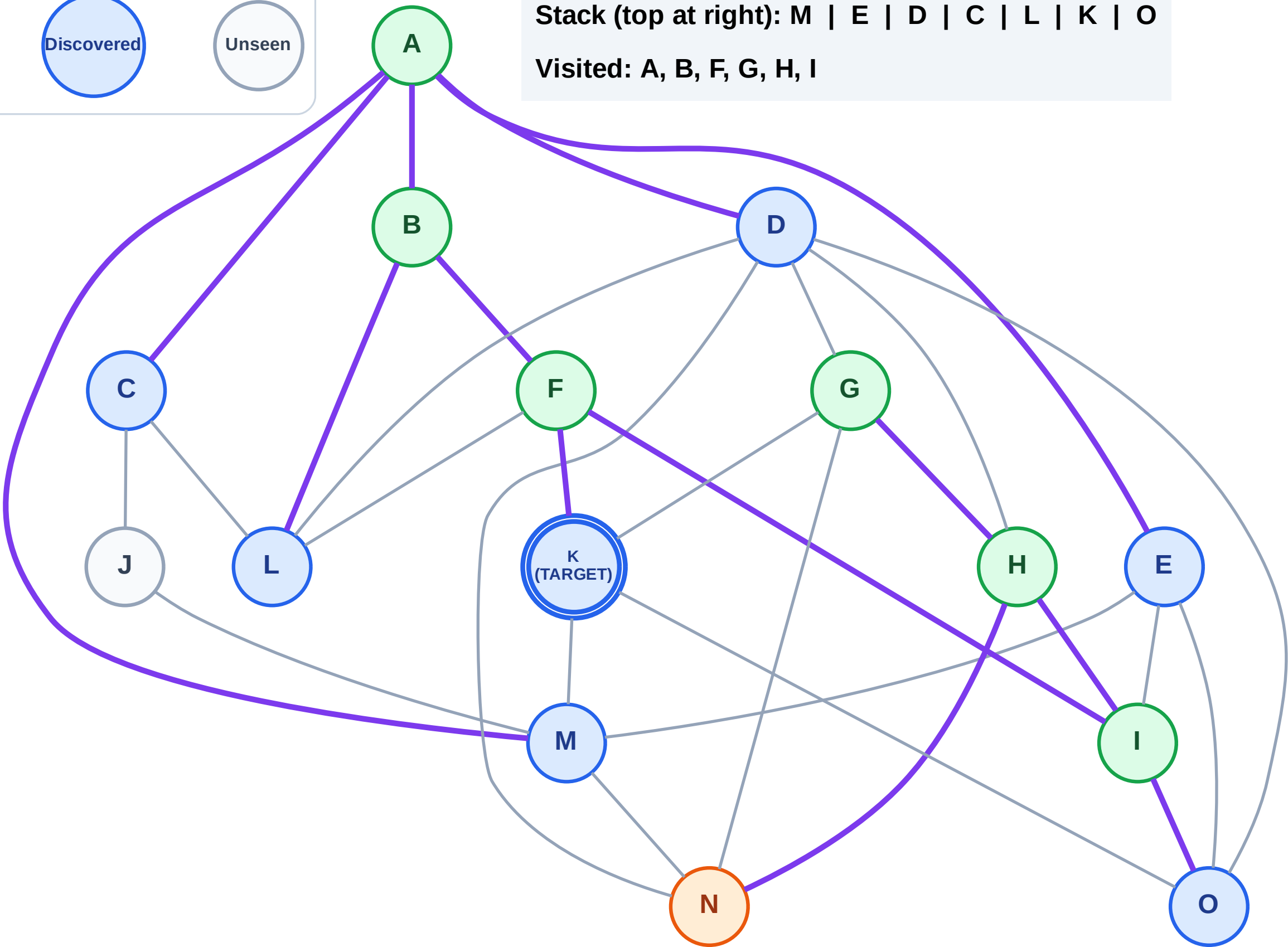
Complete

Processing

Discovered

Unseen

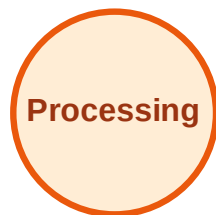
Stack (top at right): M | E | D | C | L | K | O  
Visited: A, B, F, G, H, I



# DFS Traversal

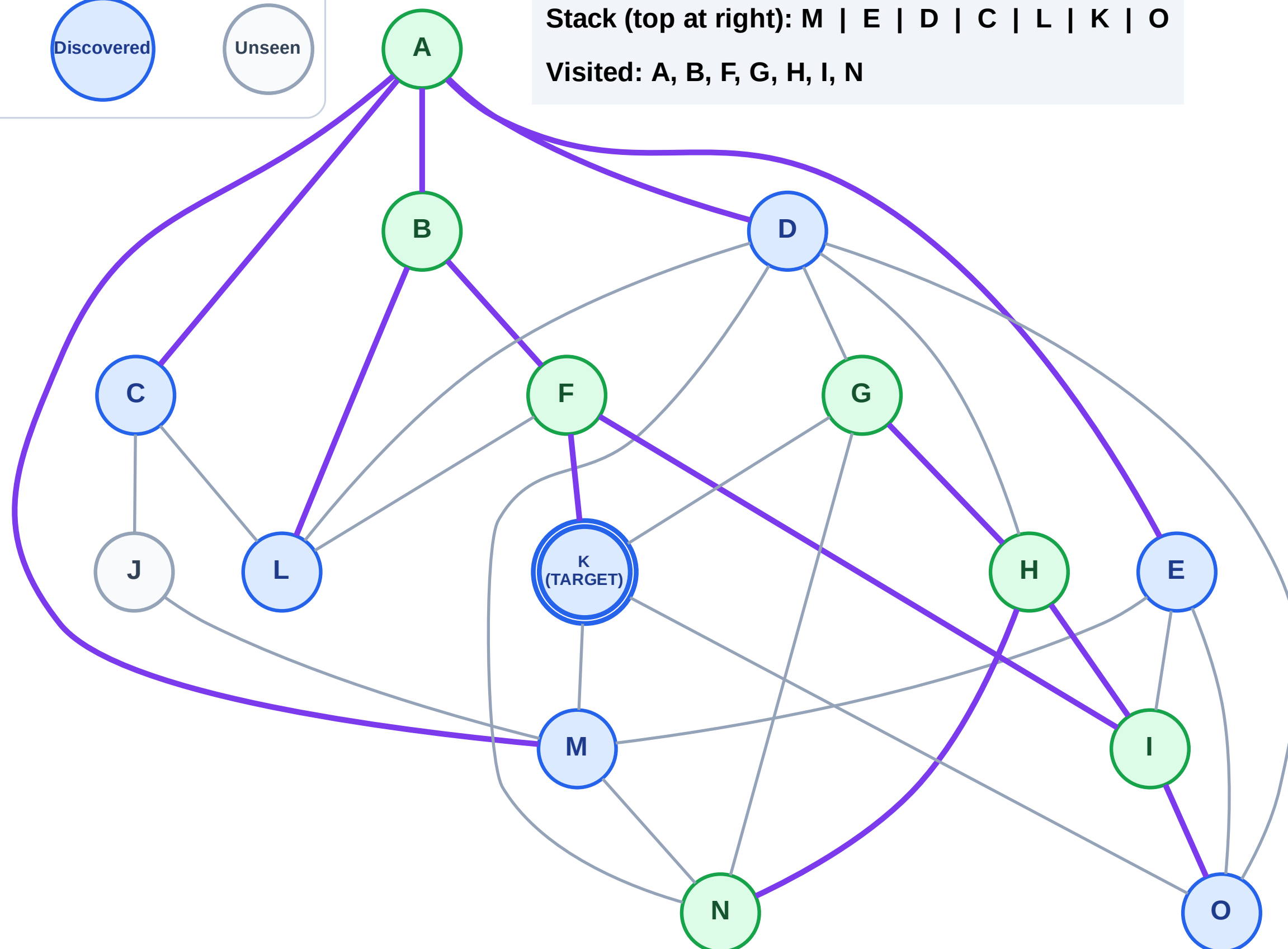
Step 15: No new neighbors from N

## Legend



Stack (top at right): M | E | D | C | L | K | O

Visited: A, B, F, G, H, I, N



# DFS Traversal

Step 16: Process node O

Legend

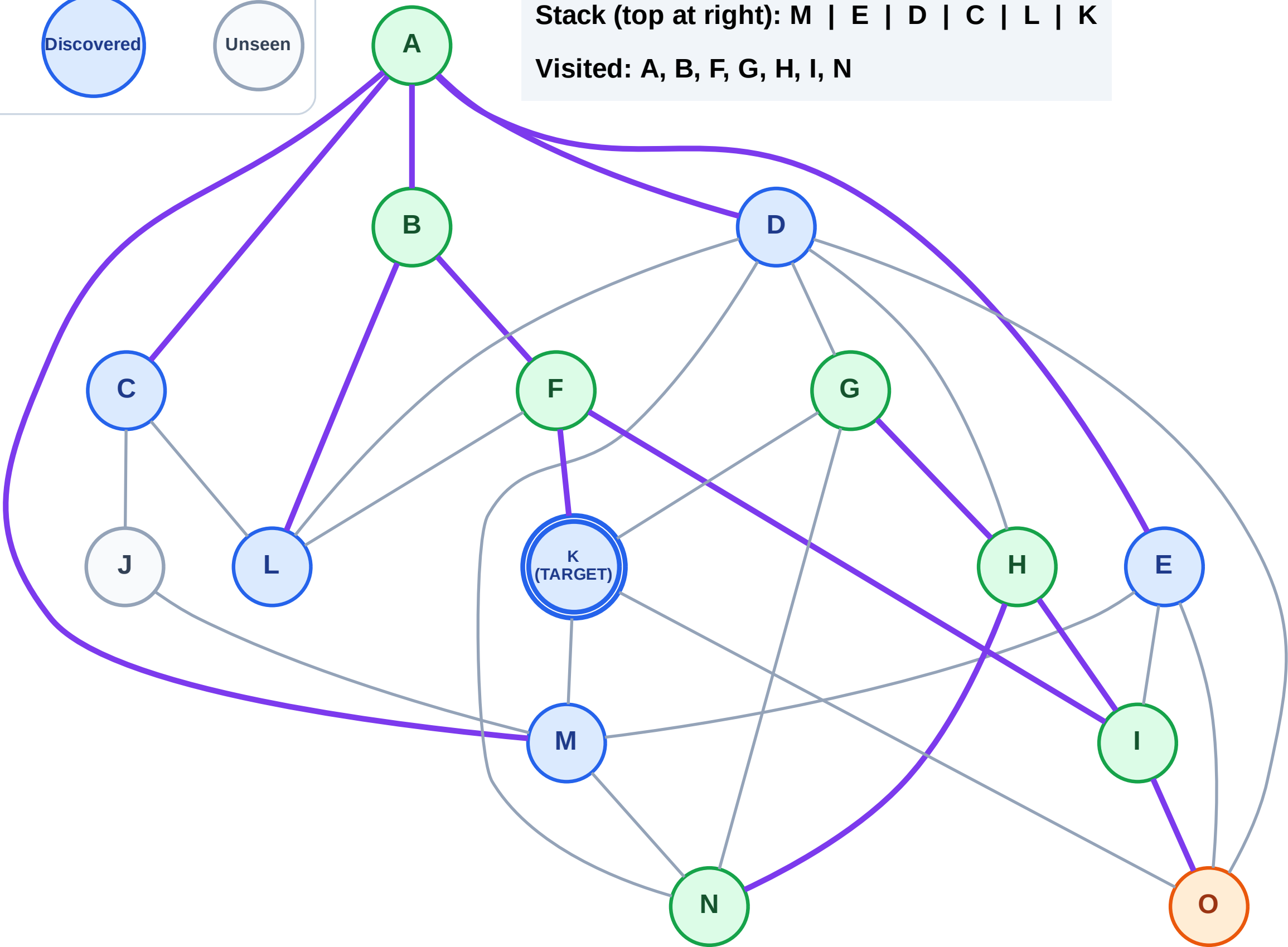
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K  
Visited: A, B, F, G, H, I, N



# DFS Traversal

Step 17: No new neighbors from O

Legend

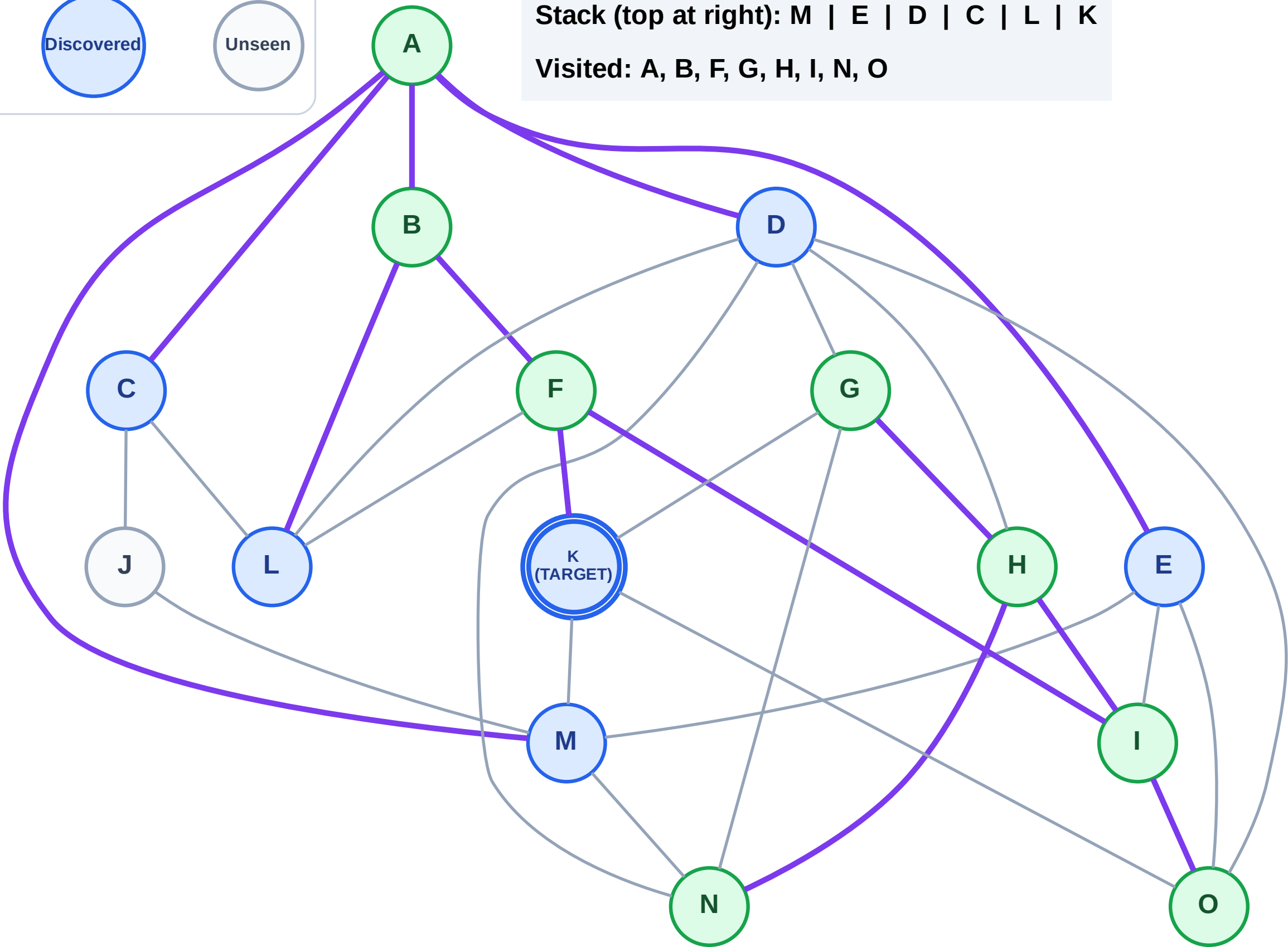
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K  
Visited: A, B, F, G, H, I, N, O



# DFS Traversal

Step 18: Process node K

Legend

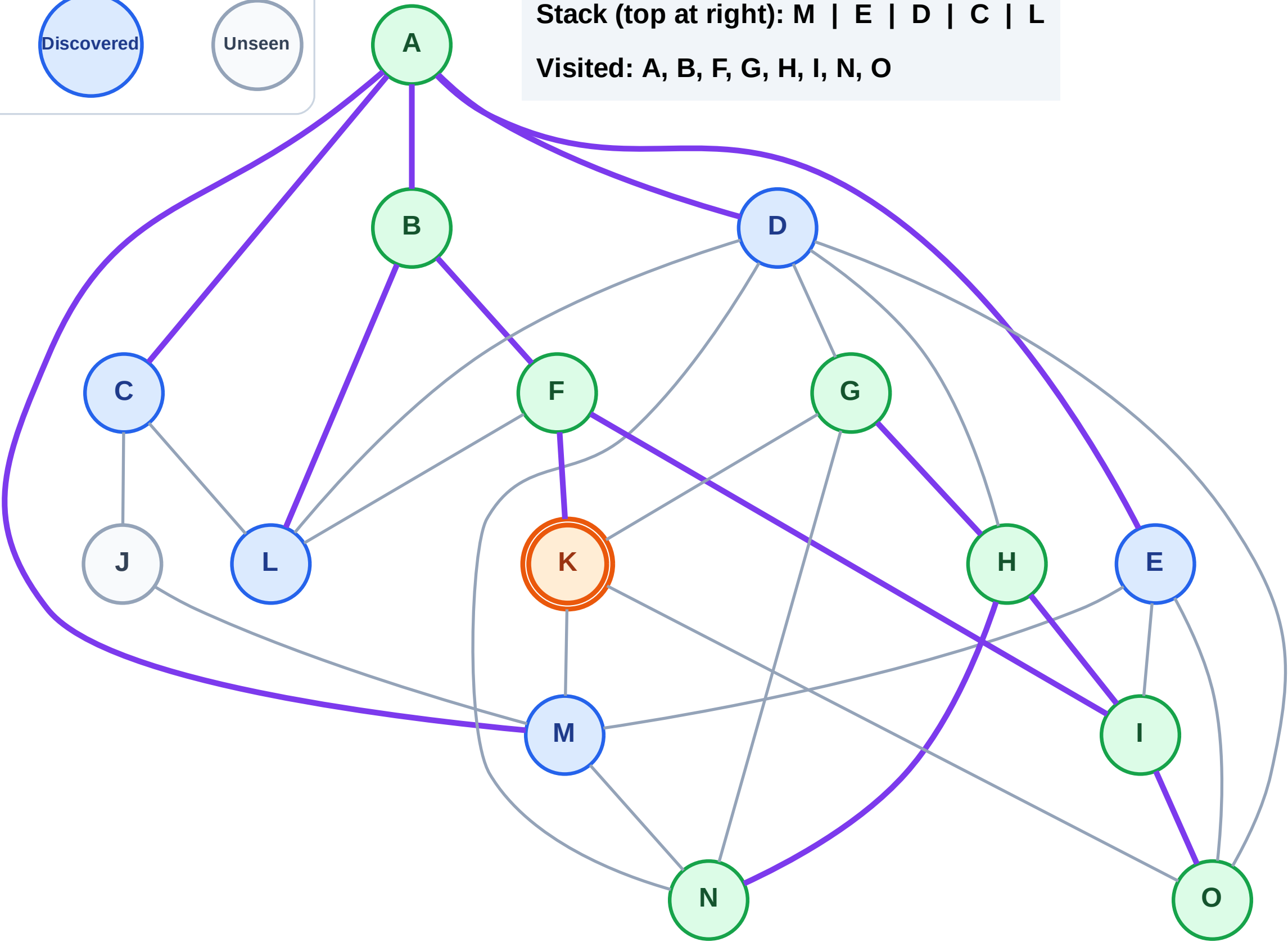
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L  
Visited: A, B, F, G, H, I, N, O



# DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L  
Visited: A, B, F, G, H, I, K, N, O

